



US012413378B1

(12) **United States Patent**
Viorel et al.

(10) **Patent No.:** **US 12,413,378 B1**
(45) **Date of Patent:** **Sep. 9, 2025**

(54) **SYSTEMS AND METHODS FOR VARIABLE TIME DIVISION DUPLEX DATA ALLOCATION**

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(21) Appl. No.: **18/066,183**

Primary Examiner — Sharad Rampuria

(22) Filed: **Dec. 14, 2022**

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Related U.S. Application Data

(63) Continuation-in-part of application No. 18/055,803,
filed on Nov. 15, 2022.

(60) Provisional application No. 63/289,521, filed on Dec.
14, 2021, provisional application No. 63/279,581,
filed on Nov. 15, 2021.

(51) **Int. Cl.**
H04L 5/14 (2006.01)
H04W 72/0457 (2023.01)
H04W 72/541 (2023.01)

(52) **U.S. Cl.**
CPC **H04L 5/1469** (2013.01); **H04W 72/0457**
(2023.01); **H04W 72/541** (2023.01)

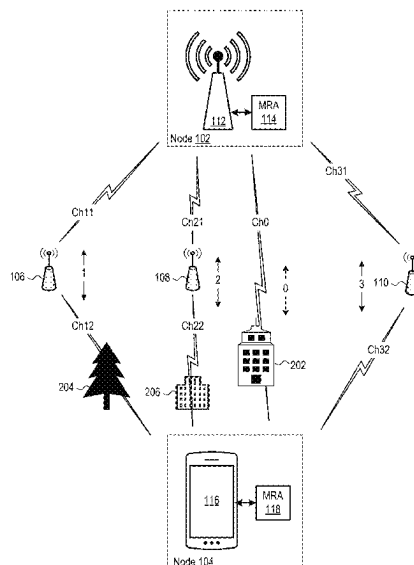
(58) **Field of Classification Search**
CPC H04L 5/1469; H04W 72/0457; H04W
72/541

See application file for complete search history.

(57) **ABSTRACT**

A method for variable time division duplex (TDD) data allocation in a communication network including a donor wireless communication node and a plurality of child wireless communication nodes. The method includes (a) determining a highest traffic child wireless communication node of the plurality of child wireless communication nodes, the highest traffic child wireless communication node having a highest data transmission requirement of the plurality of child wireless communication nodes, and (b) allocating data in first TDD data frames to be transmitted between the donor wireless communication node and a first child wireless communication node of the plurality of child wireless communication nodes according to a first data frame allocation, the first data frame allocation including allocating an on-demand portion of the first TDD data frames to data associated with the highest traffic child wireless communication node.

19 Claims, 13 Drawing Sheets



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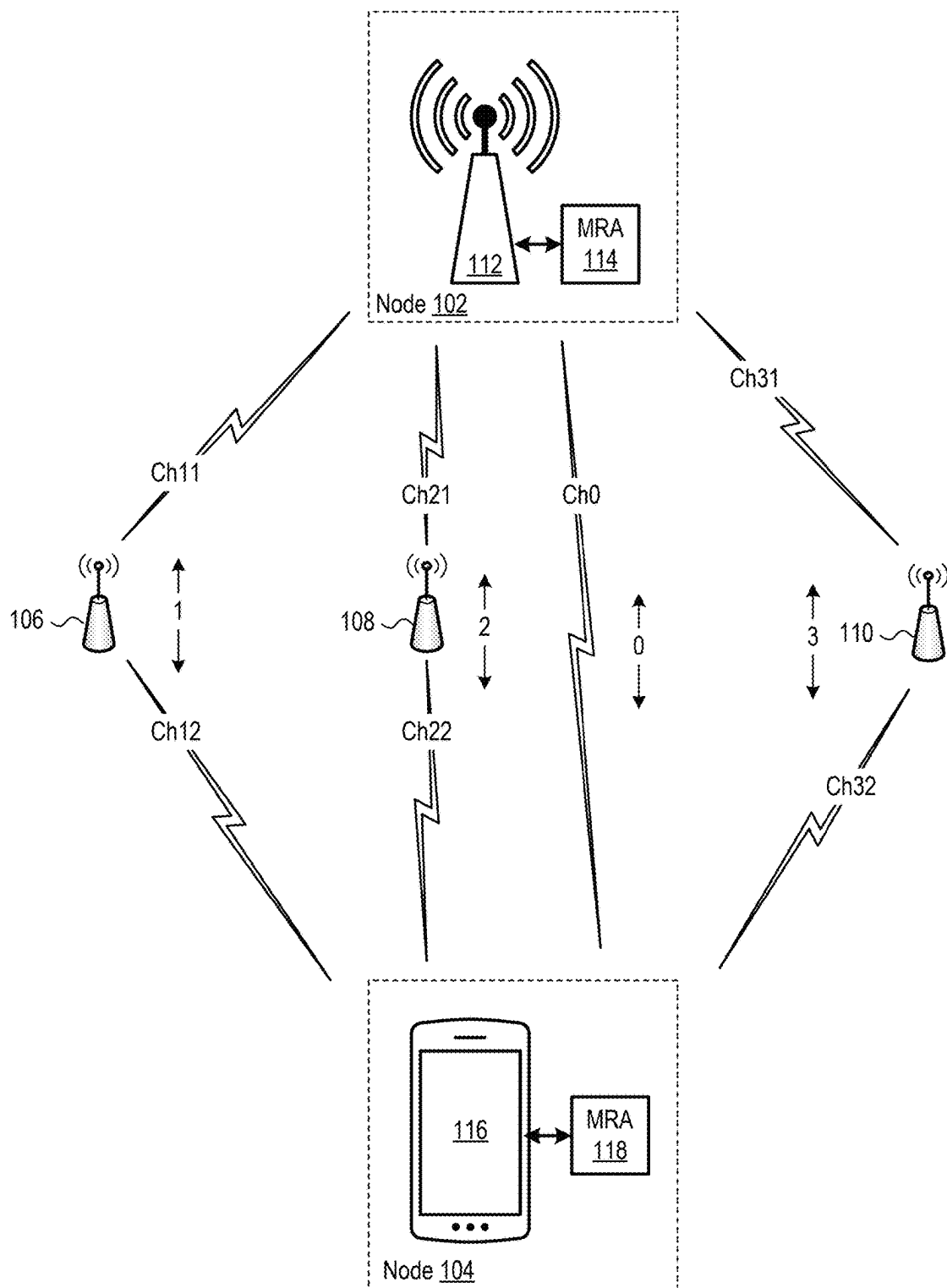


FIG. 1

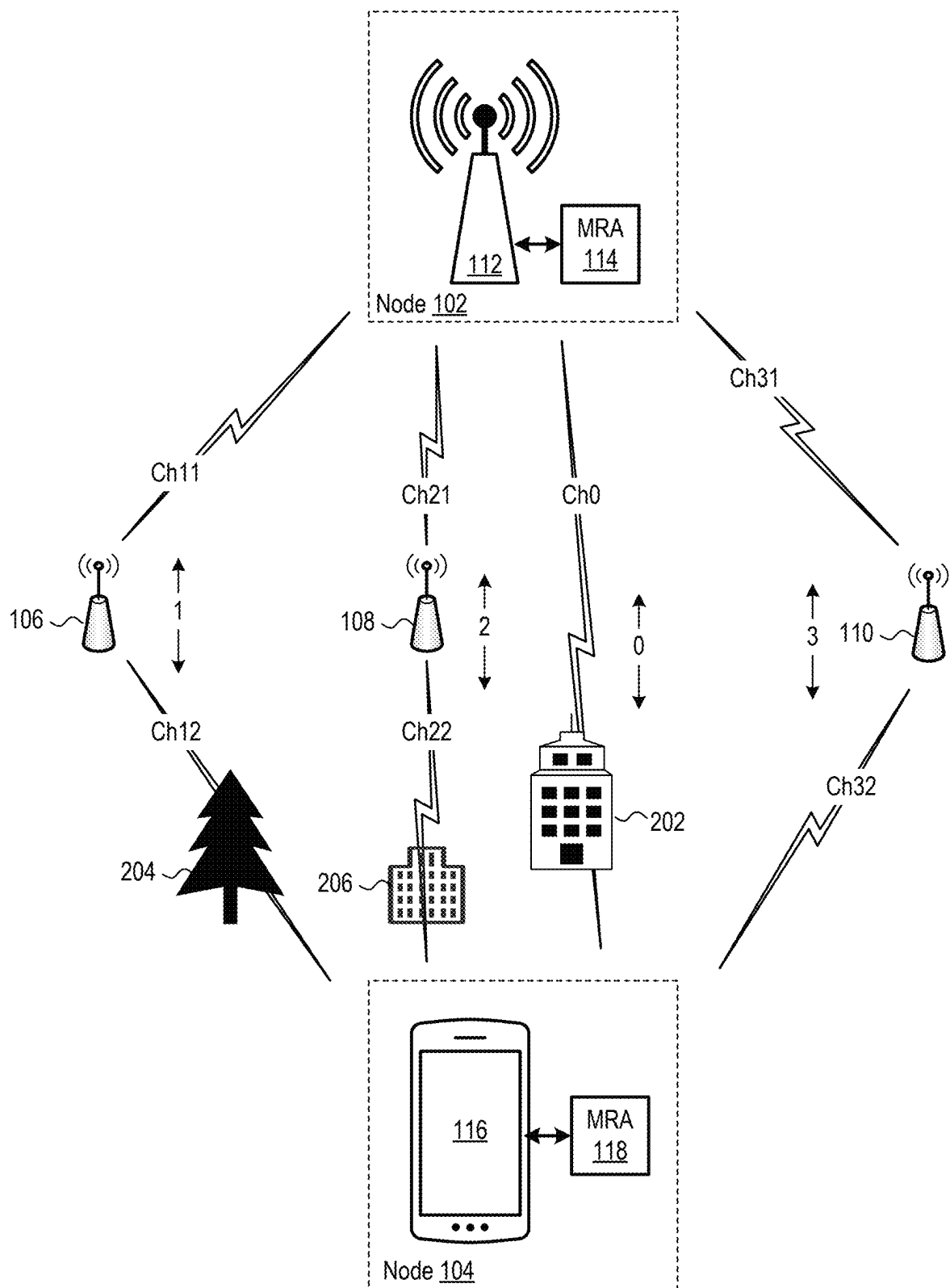
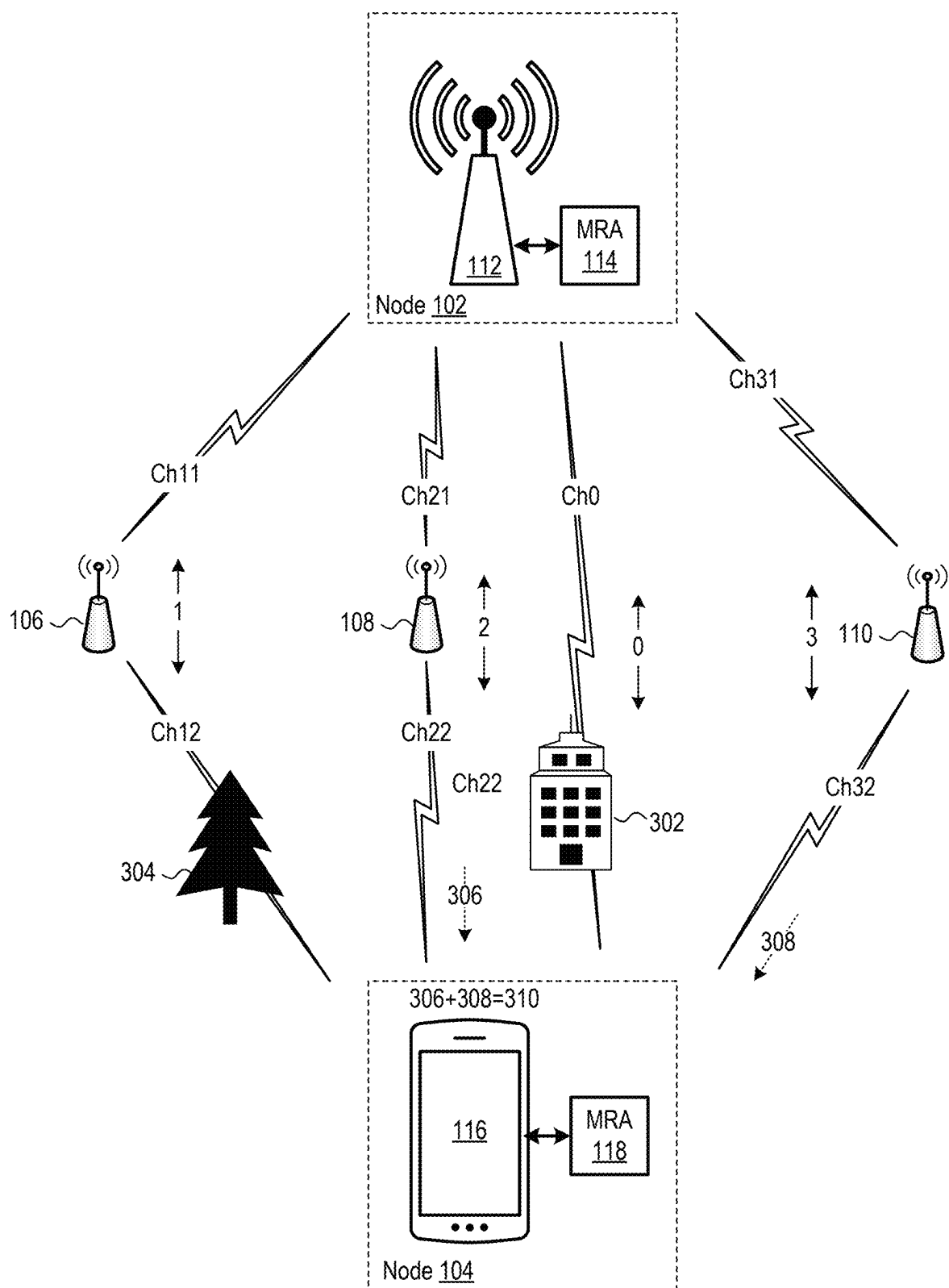


FIG. 2



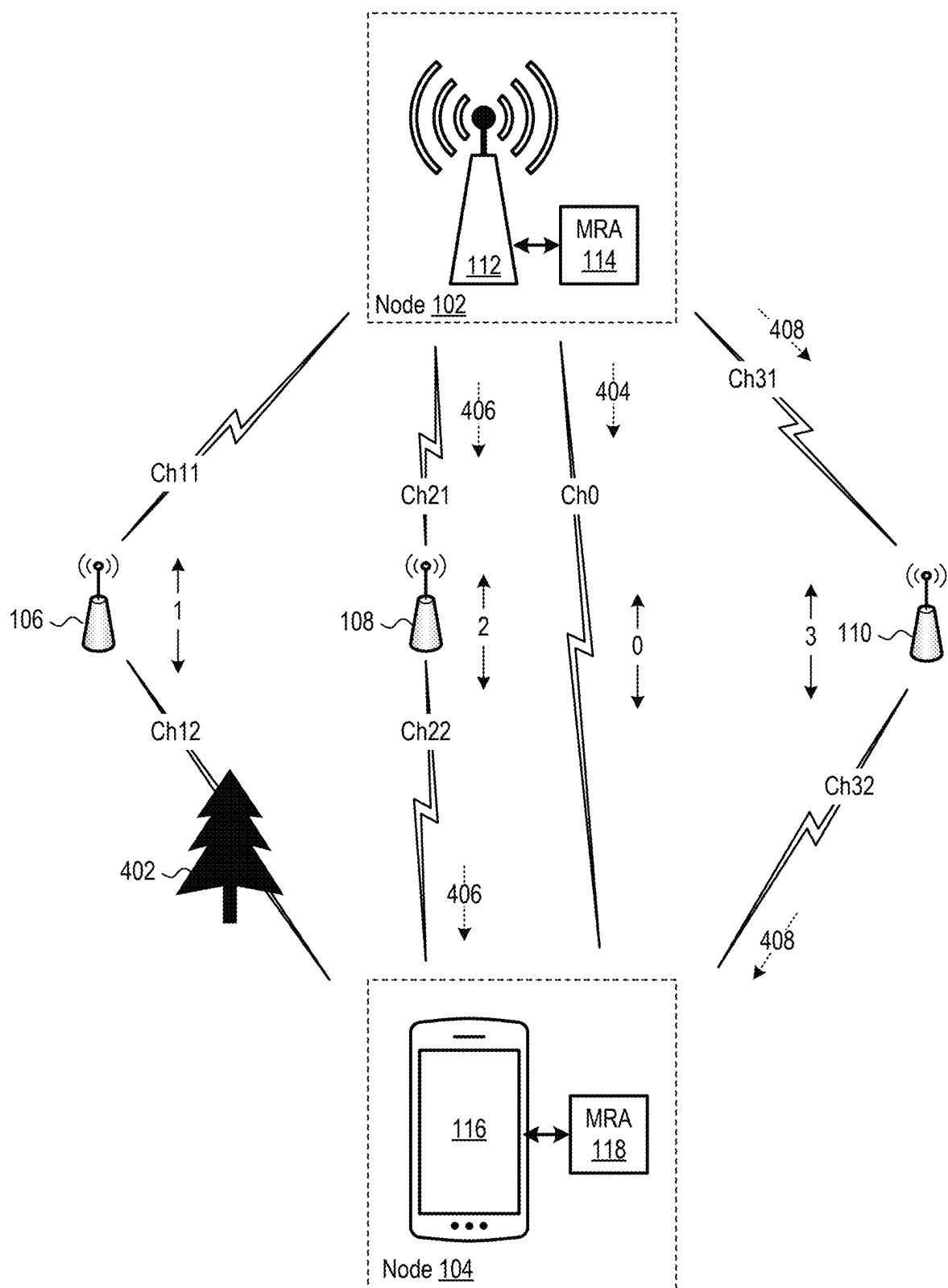
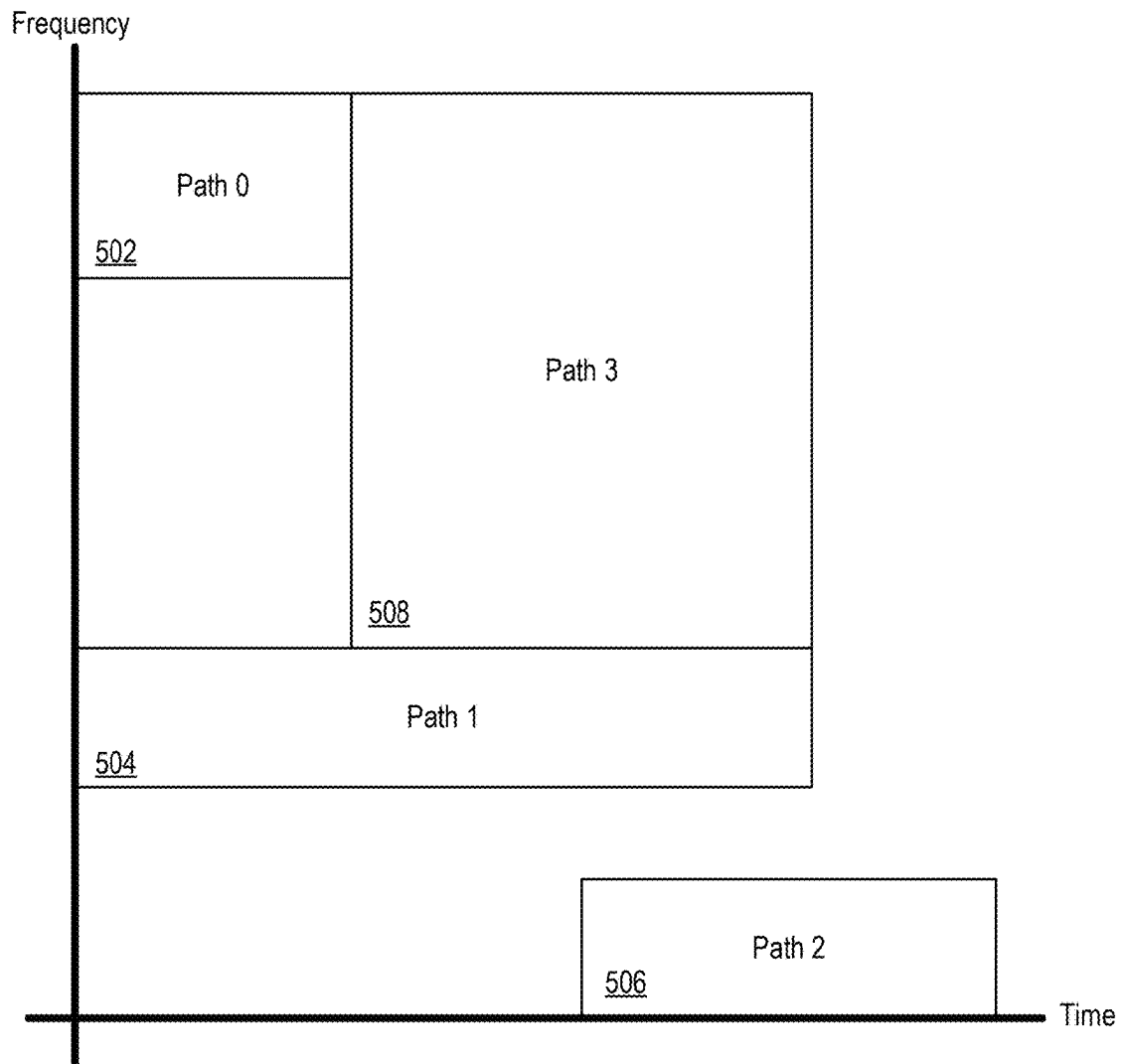


FIG. 4

500
↘**FIG. 5**

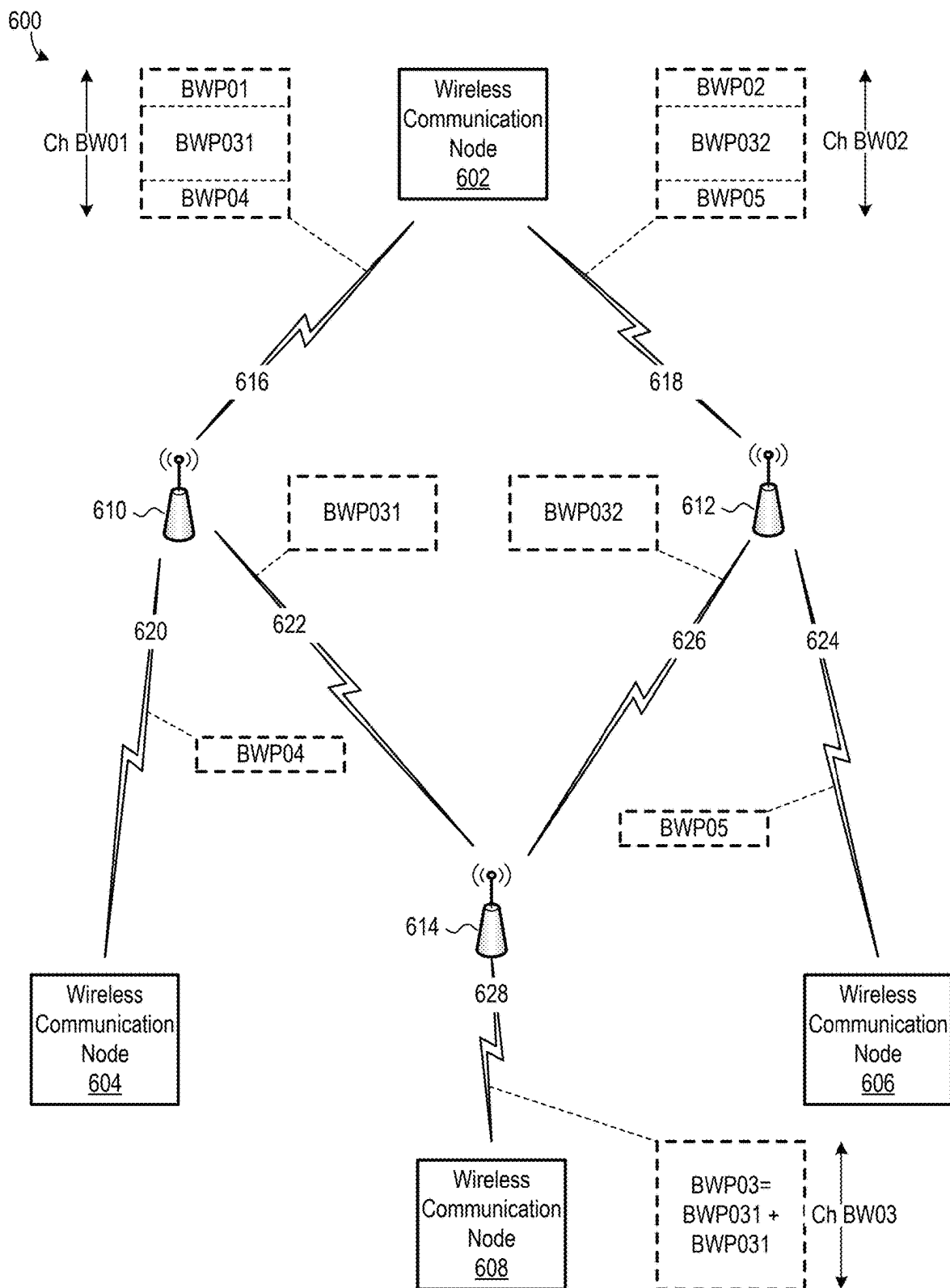


FIG. 6

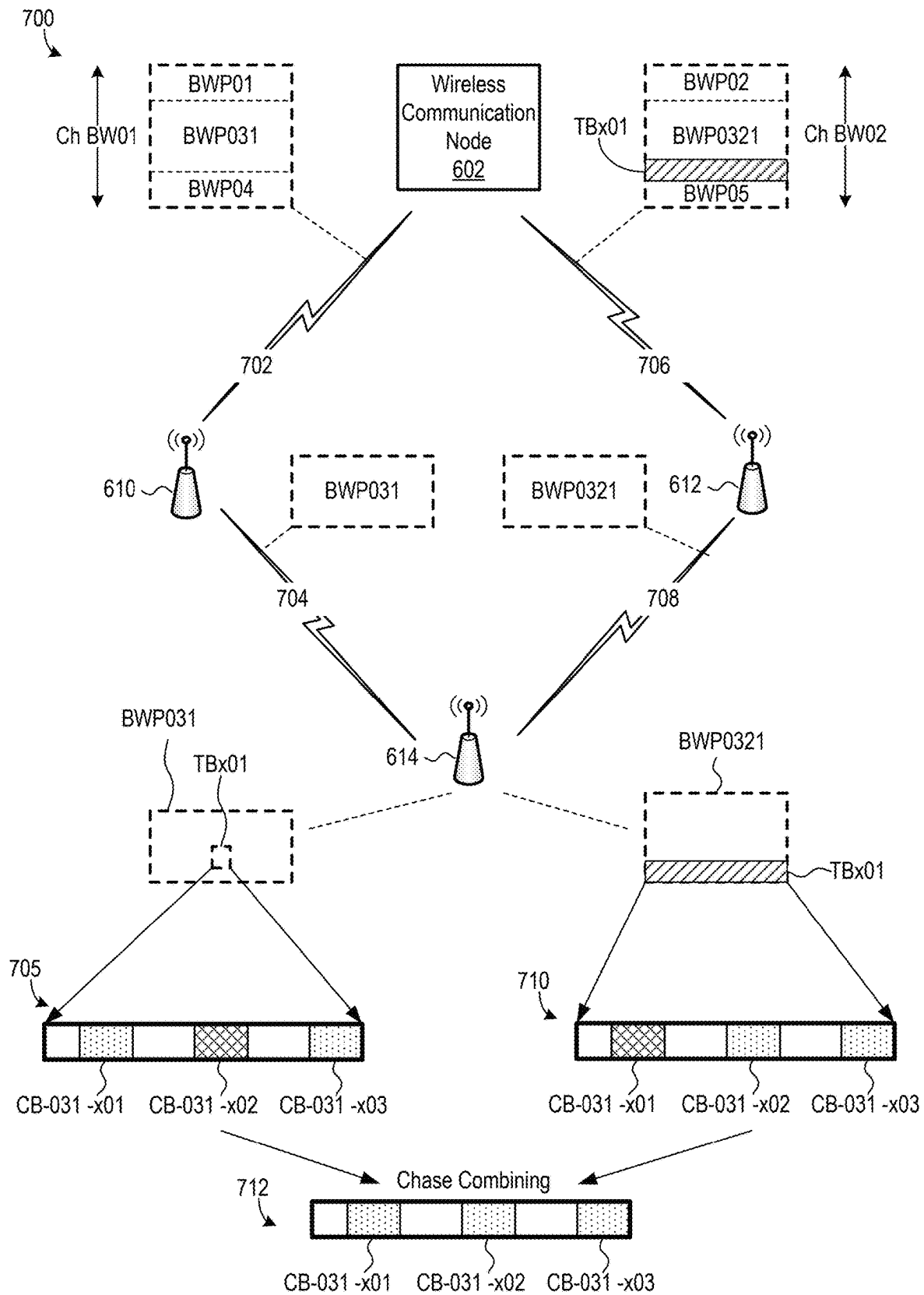


FIG. 7

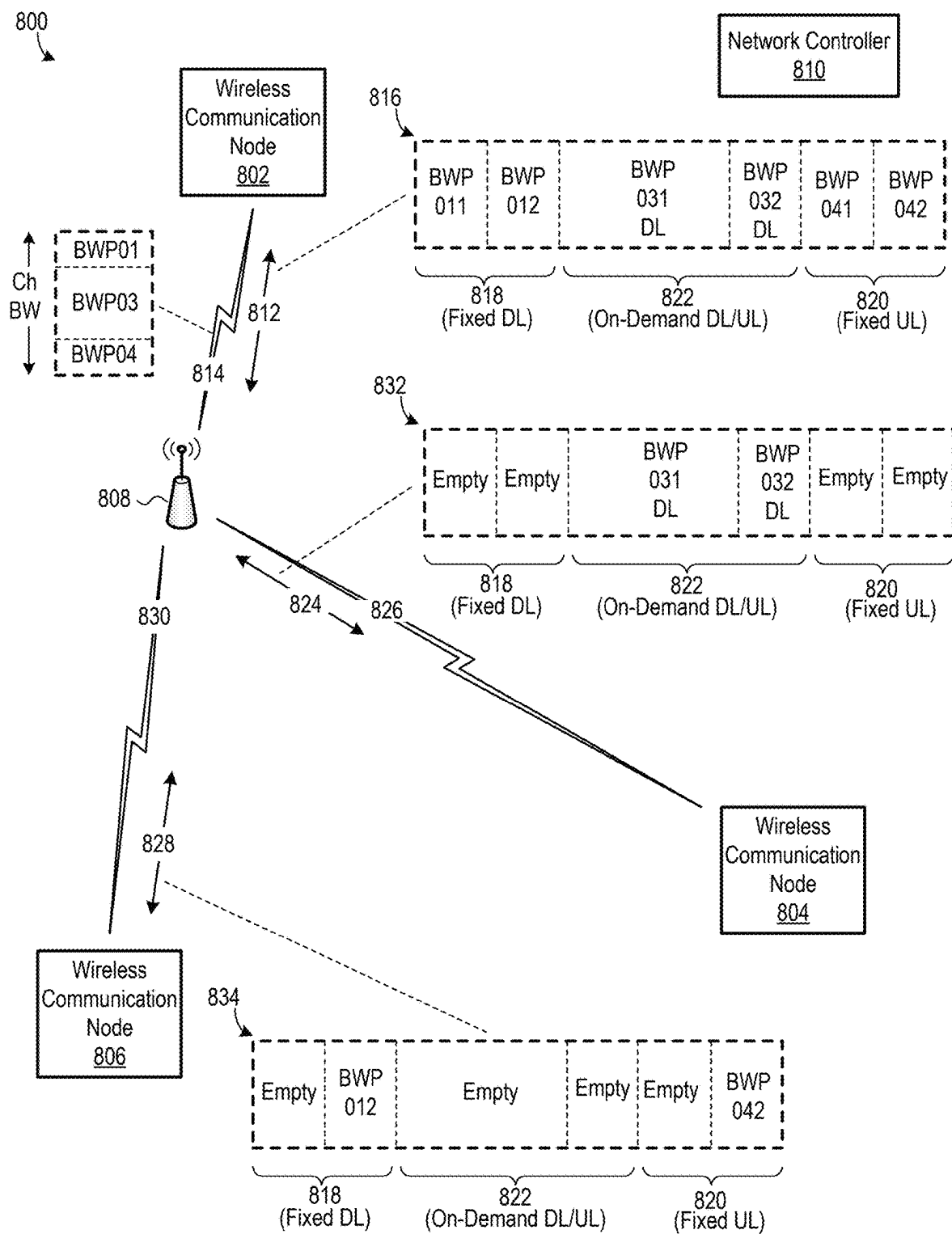


FIG. 8

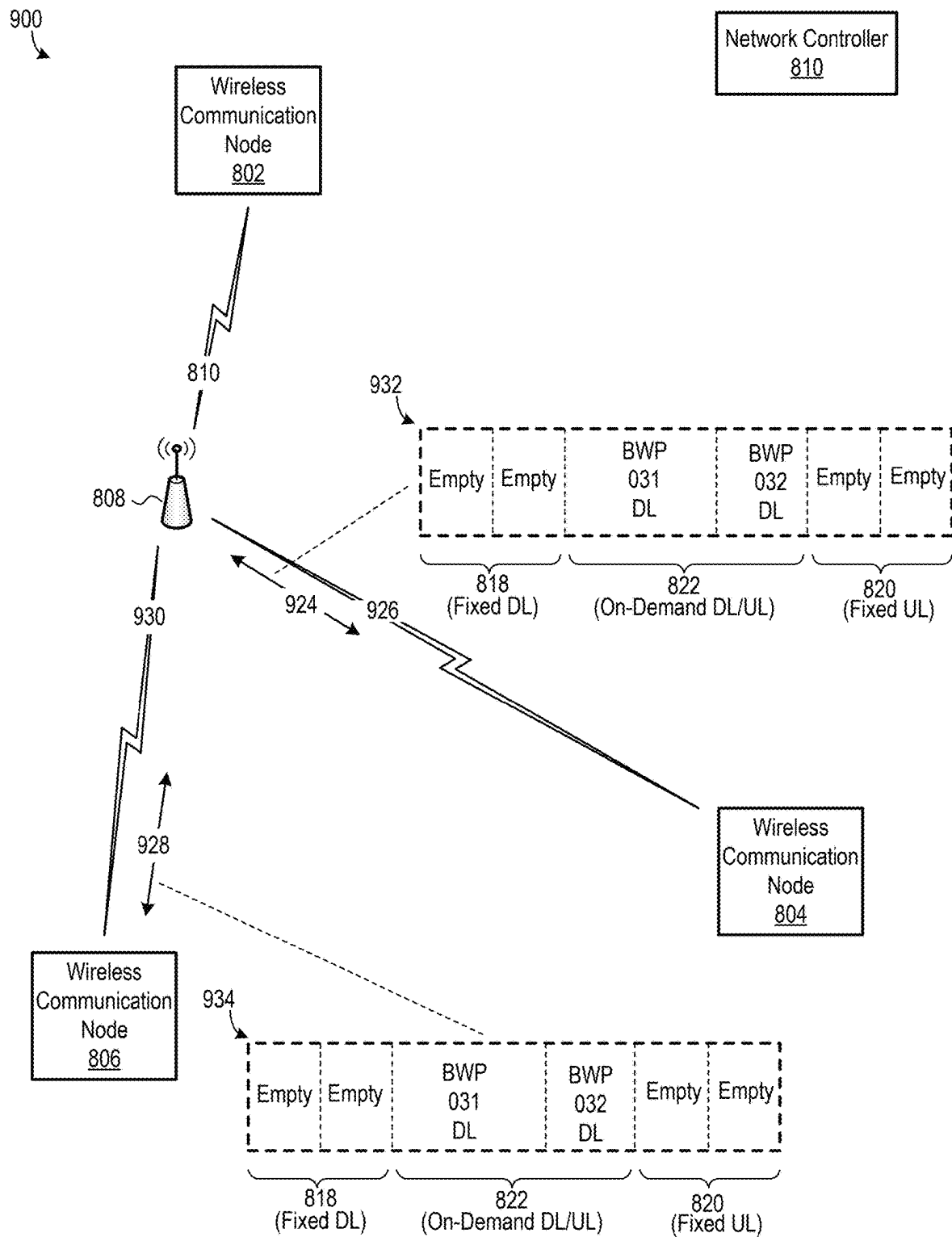


FIG. 9

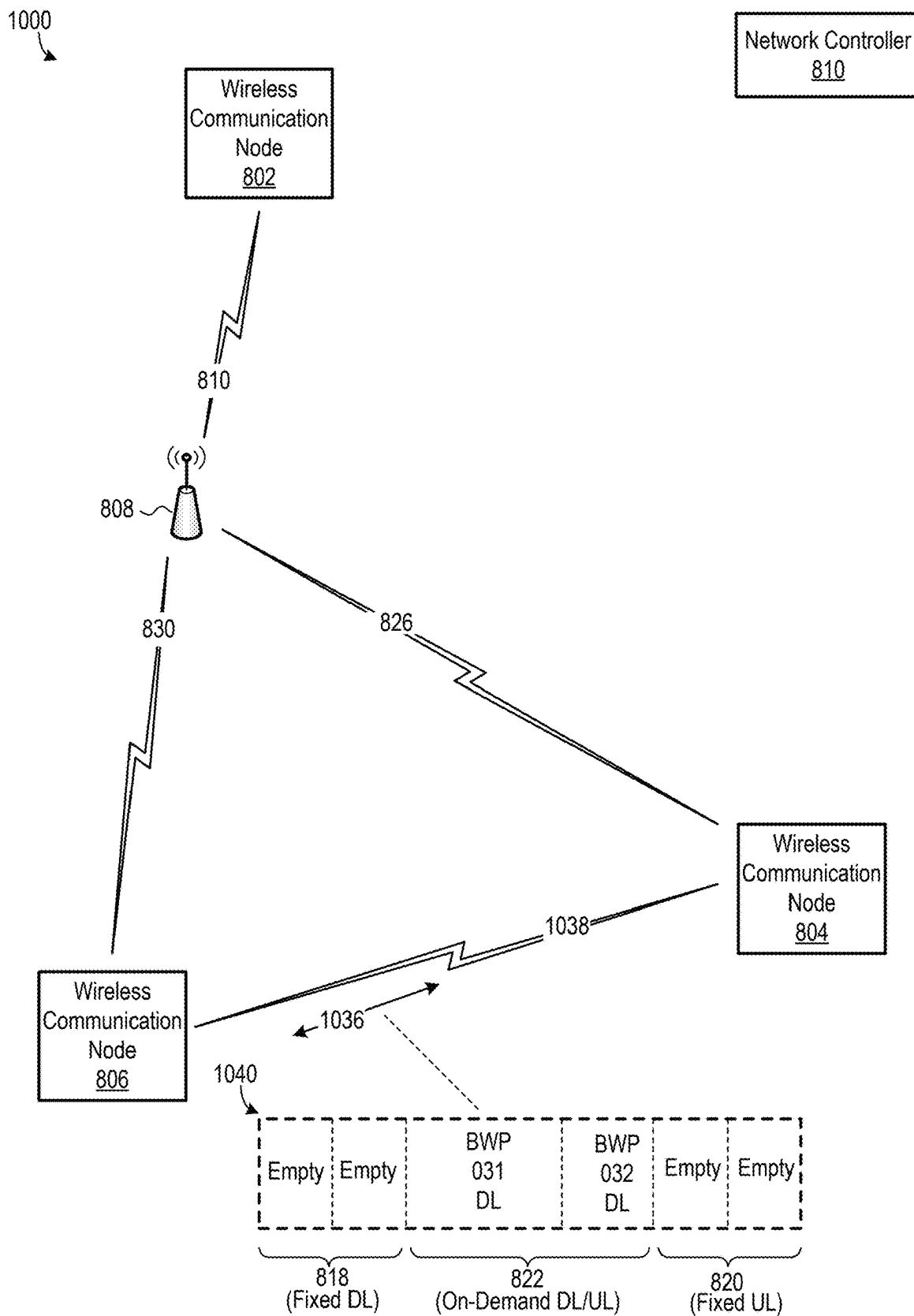


FIG. 10

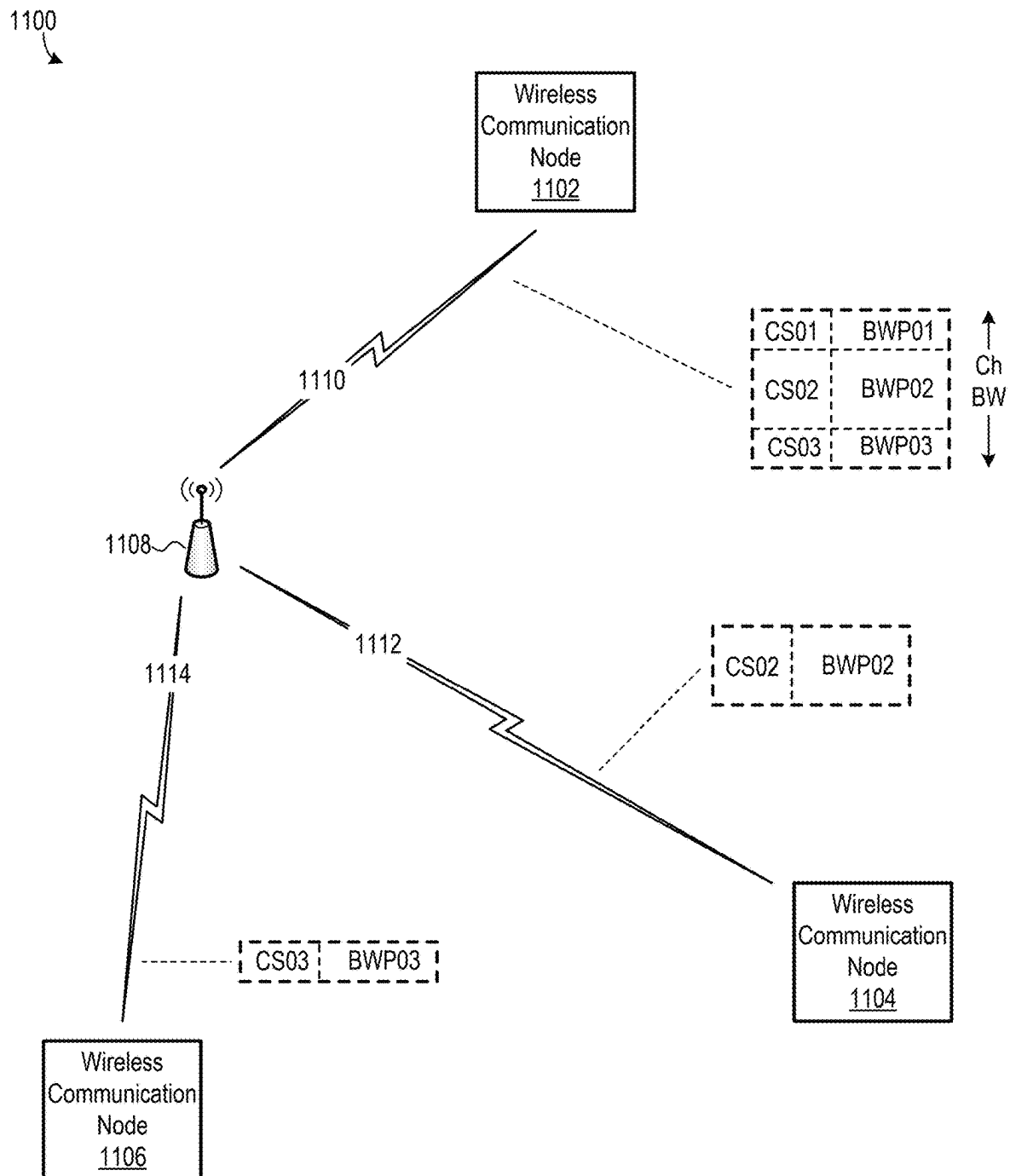


FIG. 11

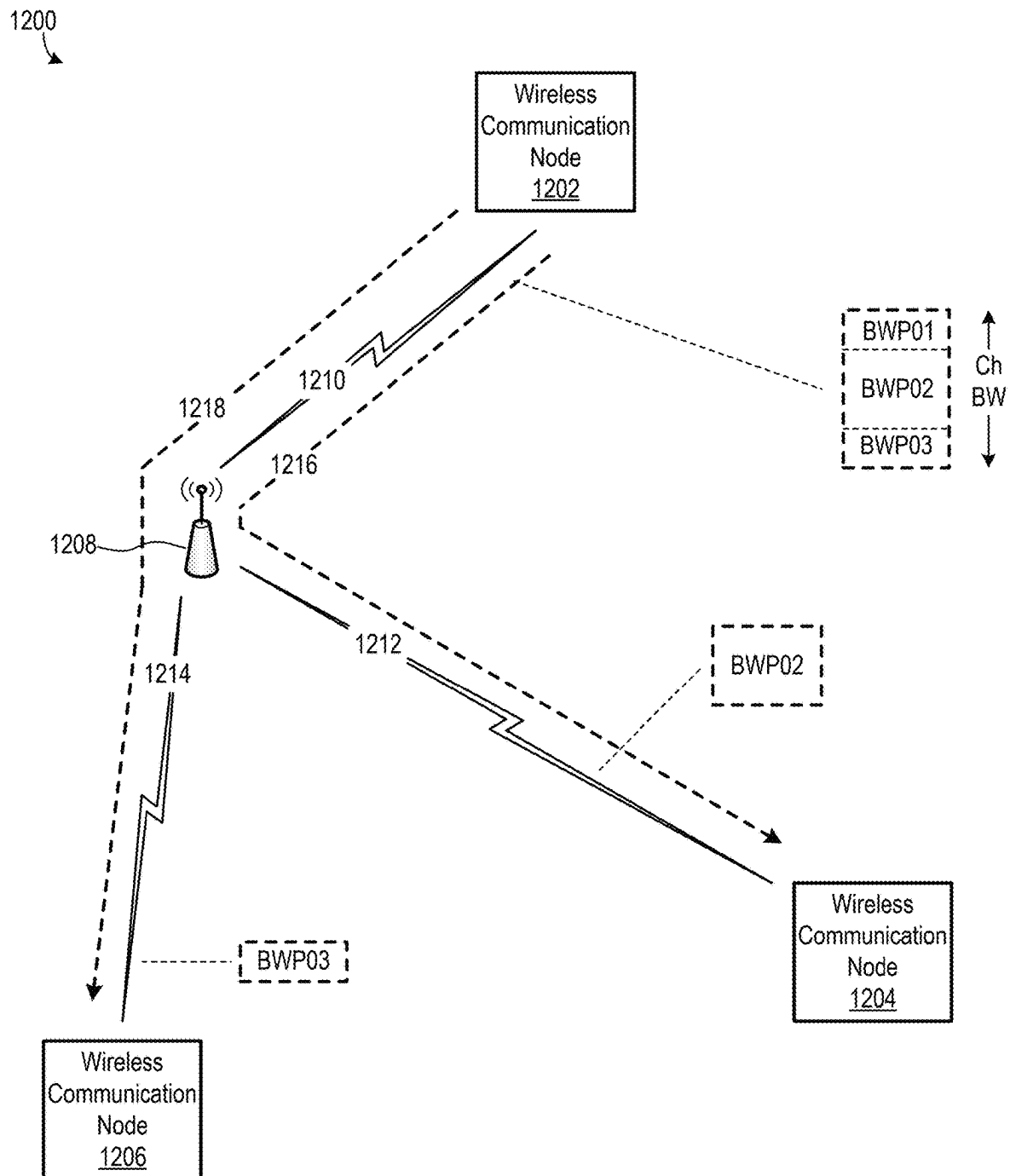
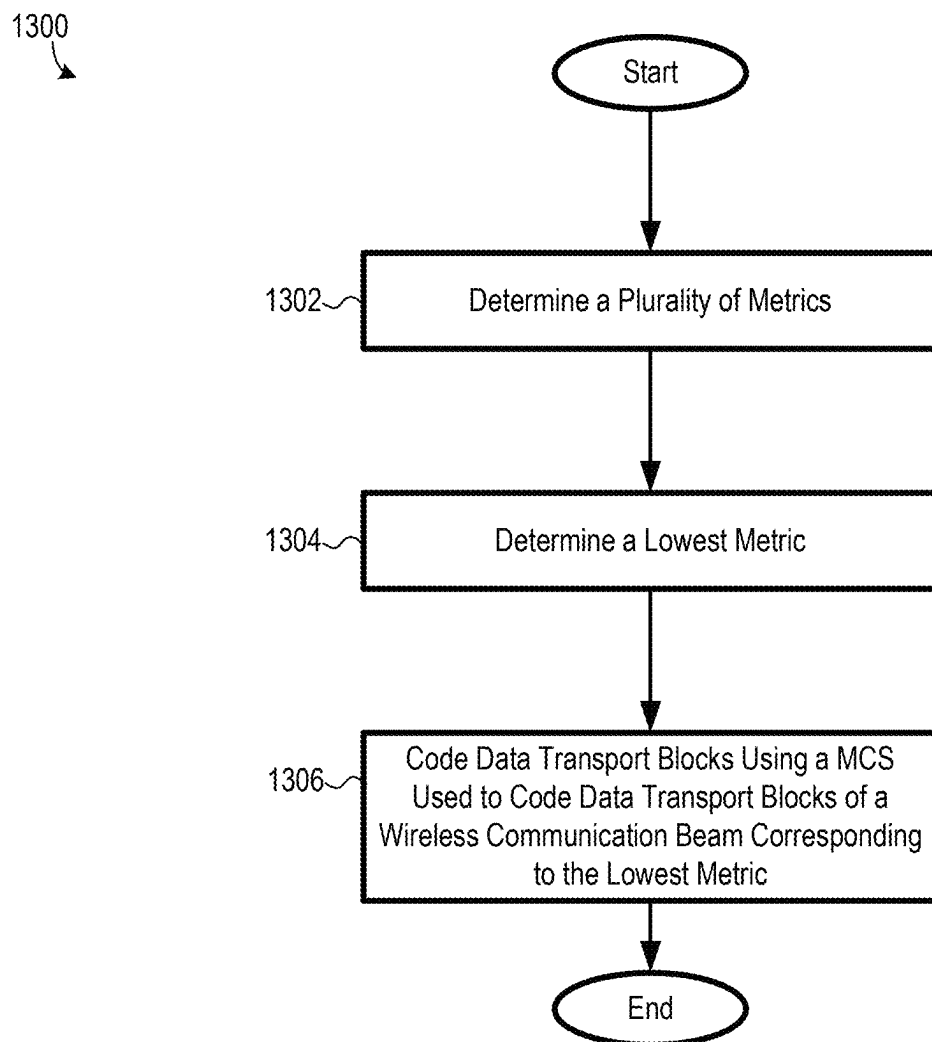


FIG. 12

**FIG. 13**

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SYSTEMS AND METHODS FOR VARIABLE TIME DIVISION DUPLEX DATA ALLOCATION

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a continuation in part of U.S. patent application Ser. No. 18/055,803, filed on Nov. 15, 2022, which claims the benefit of priority to U.S. Provisional Patent Application No. 63/279,581, filed on Nov. 15, 2021. This application also claims the benefit of priority to U.S. Provisional Patent Application No. 63/289,521, filed on Dec. 14, 2021. Each of the aforementioned patent applications is incorporated herein by reference.

BACKGROUND

Wireless communication networks transmit wireless communication signals between two or more wireless communication nodes. Examples of wireless communication nodes include, but are not limited to, wireless base stations, user equipment (UE) devices, and radio frequency (RF) repeaters. As such, performance of a wireless communication network is at least partially a function of quality of wireless communication links between wireless communication nodes.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a block diagram of a wireless communication network including a system for multiple repeater aggregation (MRA), according to an embodiment.

FIG. 2 is a block diagram illustrating an embodiment of a method for MRA including selecting a single wireless communication path to transmit data between two wireless communication nodes.

FIG. 3 is a block diagram illustrating an embodiment of a method for MRA including combining copies of a common wireless communication signal.

FIG. 4 is a block diagram illustrating an embodiment of a method for MRA including joint transmission.

FIG. 5 is a graph illustrating one example allocation of communication resources in frequency and time across wireless communication paths of the FIG. 1 wireless communication network.

FIG. 6 is a block diagram of a wireless communication network illustrating an example of a method for load balancing, according to an embodiment.

FIG. 7 is a block diagram of an embodiment of the FIG. 6 wireless communication network configured to retransmit a data structure via an alternate path, in response to the data structure including errors.

FIG. 8 is a block diagram of a wireless communication network including a system for variable time division duplex (TDD) data allocation, according to an embodiment.

FIG. 9 is a block diagram of an embodiment of the FIG. 8 wireless communication network where a wireless communication node is configured to relay local data between child wireless communication nodes, as well as to forward data between a donor wireless communication node and child wireless communication nodes.

FIG. 10 is a block diagram of an embodiment of the FIG. 8 wireless communication network where two child wireless communication nodes are configured to directly communicate with each other.

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FIG. 11 is a block diagram of a wireless communication network including a system for coreset reuse, according to an embodiment.

FIG. 12 is a block diagram of a wireless communication network including a system for bandwidth part coding, according to an embodiment.

FIG. 13 is a flow chart of a method for bandwidth part coding, according to an embodiment.

DETAILED DESCRIPTION OF THE EMBODIMENTS

A distance between two wireless communication nodes may be too long to transmit robust wireless communication signals between the two wireless communication nodes. Additionally, an obstacle, such as by a tree, a building, or a natural feature, may block direct transmission of wireless communication signals between two wireless communication nodes. Radio frequency (RF) repeaters, and similar devices, may be deployed to at least partially overcome such problems by extending wireless communication network range and/or by providing alternate paths for wireless communication signals to travel between wireless communication nodes.

A standard RF repeater, sometimes referred to as a RF relay, simply receives wireless communication signals, amplifies the wireless communication signals, and forwards or “repeats” the wireless communication signals to another wireless communication node. For example, the Third Generation Partnership Project (3GPP) specifies a RF repeater in the fifth generation (5G) new radio (NR) Release 17 as a device which works on a RF layer and simply amplifies and forwards all wireless communication signals it receives. A standard RF repeater does not have wireless communication beam management capability, and a standard RF repeater is typically omnidirectional. While a standard RF repeater supports full-duplex communication, a standard RF repeater is not capable of distinguishing between downlink data and uplink data. Additionally, a standard RF repeater amplifies noise and interference along with wireless communication signals. As such, deployment of a standard RF repeater in a wireless communication network may or may not improve signal to interference plus noise ratio (SINR) in the wireless communication network, depending on the environment and the operating conditions of the wireless communication network.

A smart RF repeater (SR), which may also be referred to as a network controlled repeater, has more functionality than a standard RF repeater. For example, the 3GPP 5G NR Release 18 specifies a smart RF repeater as a device which works on the RF layer of the user plane, and well as on the physical (PHY) and RF layers of the control plane. Release 18 also specifies that a smart repeater supports wireless communication beam management and that a smart repeater is capable of distinguishing between downlink data and uplink data.

Disclosed herein are new systems and methods which advance the state of the art of mesh wireless communication networks. Certain embodiments are operable with essentially any device which repeats wireless communication signals, including but not limited to, RF repeaters, RF relays, smart RF repeaters, and Integrated Access Backhaul (IAB) devices. Other embodiments leverage functionality of smart RF repeaters and similar devices.

The new systems and methods can be implemented in essentially any type of wireless communication network. For example, the new systems and methods could be embodied

ied in 3GPP wireless communication networks, including, but not limited to, long term evolution (LTE) wireless communication networks, fifth generation (5G) wireless communication networks, sixth generation (6G) wireless communication networks, and future developed 3GPP wireless communication networks. As another example, the new systems and methods could be embodied in non-3GPP wireless communication networks, including, but not limited to, Wi-Fi wireless communication networks, satellite wireless communication networks (e.g., using very low earth orbit (VLEO) satellites, low earth orbit (LEO) satellites, medium earth orbit (MEO) satellites, or geostationary equatorial orbit (GEO) satellites), Bluetooth wireless communication networks, long range (LoRa) wireless communication protocol, Zigbee wireless communication networks, Z-Wave wireless communication networks, etc. Furthermore, the new systems and methods could be embodied in wireless communication networks using licensed RF spectrum, unlicensed RF spectrum, or a combination of licensed and unlicensed RF spectrum.

Multiple Repeater Aggregation

Applicant has developed systems and methods for multiple repeater aggregation (MRA) which leverage multiple devices which repeat wireless communication signals, such as in a mesh wireless communication network. Certain embodiments are capable of improving wireless communication network coverage and reliability, increasing wireless communication channel capacity for downlink data and/or uplink data, and or extending wireless communication network coverage. For example, some embodiments enable a wireless communication network to maintain service to a wireless communication node even if some wireless communication signals to the wireless communication node are blocked by an obstacle, e.g., one wireless communication beam is blocked, thereby helping prevent shadowing loss. As another example, certain embodiments benefit from diversity gain, which mitigates small scale fading. As an additional example, particular embodiments are capable of serving a given wireless communication node via multiple channels with low correlation between the channels, or even with no correlation between the channels, which increases channel capacity according to Shannon theory.

While the new systems and methods for MRA may achieve advantages in essentially any application, they may be especially beneficial in fixed wireless access (FWA) applications, vehicular applications, and aircraft applications. In particular, FWA applications may be particularly challenging due to FWA customer premises equipment (CPE) frequently being located indoors, as well as due to FWA applications typically requiring more throughput than mobile applications. Use of the new MRA systems and methods may help improve coverage indoors and help achieve high throughput, which may make them particularly beneficial in FWA applications. Vehicular applications may also be particularly challenging due to vehicle movement, such as due to doppler shift resulting from vehicle movement. Use of the new MRA systems and methods may help mitigate such challenges. Furthermore, aircraft, such as drones, may be difficult to adequately serve using conventional wireless communication networks because conventional wireless communication networks are typically optimized for terrestrial users with down tilt in wireless base station antennas, causing coverage towards the sky to be poor. The new MRA systems and methods may be used in wireless communication networks to improve coverage towards the sky, which benefits aircraft applications.

FIG. 1 is a block diagram of a wireless communication network 100 which includes an embodiment of the new systems for MRA. Wireless communication network 100 includes a wireless communication node 102, a wireless communication node 104, a RF repeater 106, a RF repeater 108, and a RF repeater 110. It is understood, however, that wireless communication network 100 can include any quantity of wireless communication nodes and any quantity of RF repeaters, as long as wireless communication network 100 includes at least two wireless communication nodes and at least two RF repeaters. Particular embodiment of the new systems and methods for MRA do not require coordination among devices which repeat wireless communication signals. As such, in certain embodiments, RF repeaters 106, 108, and 110 are conventional RF repeaters or conventional RF relays. However, RF repeaters 106, 108, and 110 could be more sophisticated devices, such as smart RF relays or IAB devices, as well as custom devices, without departing from the scope hereof.

Each of wireless communication node 102 and wireless communication node 104 is configured to transmit and receive wireless communication signals. Wireless communication node 102 includes a wireless base station 112 and a MRA module 114. In some embodiments, wireless base station 112 includes a 3GPP wireless base station, including, but not limited to, a NodeB, an eNode B, a gNodeB, or a future-developed 3GPP wireless base station. In some other embodiments, wireless base station 112 is a non-3GPP wireless base station, including, but not limited to, a Wi-Fi wireless base station, a satellite wireless base station, a Bluetooth wireless base station, a LoRa wireless base station, a Zigbee wireless base station, a Z-Wave wireless base station, etc. It should be noted that wireless base station 112 need not be a complete wireless base station. For example, in particular embodiments, wireless base station 112 is a remote radio head of a wireless base station. Additionally, wireless base station 112 could be replaced with a different type of device capable of transmitting and receiving wireless communication signals, such as a RF repeater or a user equipment (UE) device.

Wireless communication node 104 includes a UE device 116 and a MRA module 118. Although UE device 116 is depicted as being a mobile phone, UE device 116 could be another type of UE device. For example, UE device 116 could alternately be a computer, a set-top device, a data storage device, an Internet of Things (IoT) device, an entertainment device, a computer networking device, a modem, a smartwatch, a wearable device with wireless capability, a medical device, a security device, a monitoring device, or a wireless access device (including, for example, an eNB, a gNB, a Wi-Fi-based wireless access point, an IAB access point, a microcell, a picocell, a femtocell, a macrocell, a Wi-Fi-based application, a satellite communication device, etc.). Additionally, UE device 116 could be replaced with a different type of device capable of transmitting and receiving wireless communication signals, such as a RF repeater or a wireless base station.

Wireless communication network 100 includes four wireless communication paths, i.e., wireless communication path 0, wireless communication path 1, wireless communication path 2, and wireless communication path 3, between wireless communication nodes 102 and 104. Wireless communication path 0 is a direct path between wireless communication node 102 and wireless communication node 104 via a wireless channel Ch0. Accordingly, wireless communication path 0 does not include a RF repeater. Each of wireless communication paths 1, 2, and 3 includes a RF repeater.

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Specifically, wireless communication path 1 includes (i) a wireless channel Ch11 between wireless communication node 102 and RF repeater 106 and (ii) a wireless channel Ch12 between RF repeater 106 and wireless communication node 104. Wireless communication path 2 includes (i) a wireless channel Ch21 between wireless communication node 102 and RF repeater 108 and (ii) a wireless channel Ch22 between RF repeater 108 and wireless communication node 104. Wireless communication path 3 includes (i) a wireless channel Ch31 between wireless communication node 102 and RF repeater 110 and (ii) a wireless channel Ch32 between RF repeater 110 and wireless communication node 104. Any of wireless communication paths 0, 1, 2, and 3 could include a plurality of wireless communication beams, and each wireless communication beam could include a plurality of streams. Any one of wireless communication paths 0, 1, 2, and 3 could carry downlink data, uplink data, or a combination of downlink data and uplink data.

It is understood that the number of wireless communication paths between wireless communication nodes 102 and 104 will vary with the number of RF repeaters in wireless communication network 100. Additionally, in some embodiments, one or more wireless communication paths between wireless communication nodes 102 and 104 may span two or more RF repeaters. Furthermore, a given wireless communication path between wireless communication nodes 102 and 104 may not be available due to, for example, the path being blocked by an obstacle.

MRA module 114 and MRA module 118 collectively form an embodiment of the new systems for MRA. In some alternate embodiments, however, one of MRA module 114 and MRA module 118 is omitted, such that solely MRA module 114 or solely MRA module 118 forms the system for MRA. Each of MRA modules 114 and 118 is implemented, for example, by a processor (not shown) executing instructions in the form of software and/or firmware stored in a data store (not shown). In some embodiment, MRA modules 114 and 118 are at least partially implemented by electronic circuitry (not shown). While MRA module 114 is depicted as being separate from wireless base station 112, MRA module 114 may be partially or fully integrated with wireless base station 112. For example, in certain embodiments, MRA module 114 is implemented by a processor of wireless base station 112 executing instructions in the form of software and/or firmware stored in a data store (e.g., a memory or a hard drive) of wireless base station 112. Similarly, although MRA module 118 is depicted as being separate from UE device 116, MRA module 118 may be partially or fully integrated with UE device 116. For example, in certain embodiments, MRA module 118 is implemented by a processor of UE device 116 executing instructions in the form of software and/or firmware stored in a data store, e.g., a memory, of UE device 116.

The system for MRA of wireless communication network 100 is configured to execute one or more of the following methods for MRA, to promote high performance of wireless communication network 100: (a) Method A—select a single path, (b) Method B—combine copies of a signal, (c) Method C—joint transmission, (d) Method D—dynamic beam configuration, (e) Method E—beam direction, (f) Method F—multiple streams, (g) Method G—frequency/time resource aggregation, (h) Method H—fake base station detection, and (i) Method I—dynamic power control. Method A—Select a Single Path

In method A, each RF repeater 106, 108, and 110 amplifies and repeats a received downlink or uplink wireless

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communication signal. Additionally, each RF repeater 106, 108, and 110 may receive the same copy of a downlink wireless communication signal or an uplink wireless communication signal sent by wireless base station 112 or UE device 116, respectively. As such, there may be multiple wireless communication paths for the same copy of a wireless communication signal between wireless communication node 102 and wireless communication node 104. However, these multiple wireless communication paths may not be equal. For example, one RF repeater 106, 108, and 110 may be blocked while another RF repeater 106, 108, and 110 may not be blocked. In method A, MRA module 114 and/or MRA module 118 select a single wireless communication path to transmit data between wireless communication node 102 and wireless communication node 104, such as based on one or more measured conditions of the wireless communication paths, to help optimize wireless communication between the two nodes.

In particular, one or more devices of wireless communication network 100 measure conditions of one or more wireless communication paths, or portions of wireless communication paths, of wireless communication network 100, and the devices send data representing the measured conditions to MRA module 114 and/or MRA module 118. For example, wireless base station 112 may measure conditions of wireless channels Ch0, Ch11, Ch21, and Ch31 and send data representing the measured conditions to MRA module 114 and/or MRA module 118. As another example, UE device 116 may measure conditions of wireless channels Ch0, Ch12, Ch22, and Ch32 and send data representing the measured conditions to MRA module 114 and/or MRA module 118. As additional example, RF repeater 106 may measure conditions of wireless channels Ch11 and Ch12 and send data representing the measured conditions to MRA module 114 and/or MRA module 118. Examples of possible measured conditions of wireless communication paths include, but are not limited to, one or more of reference signal received power (RSRP) of each wireless communication path, reference signal received quality (RSRQ) of each wireless communication path, time advance of each wireless communication path, cell ID of each wireless communication path, wireless communication beam ID of each wireless communication path, percentage of utilization of each wireless communication path, etc.

MRA module 114, MRA module 118, or the combination of MRA module 114 and MRA module 118, select one wireless communication path to transmit data between wireless communication node 102 and wireless communication node 104 at least partially based on the data representing the measured conditions. For example, MRA module 114 and/or MRA module 118 may select whichever wireless communication path 1, 2, 3, or 4 which has a highest RSRP, a highest RSRQ, a smallest time advance, a smallest percentage of channel utilization, etc. MRA module 114 and/or MRA module 118 then cause data to be routed between wireless communication node 102 and wireless communication node 104 solely via the one selected wireless communication path, such as by controlling wireless base station 112 and/or UE device 116 to route data accordingly. It should be noted that particular embodiments of Method A do not require coordination of RF repeater 106, RF repeater 108, or RF repeater 110, thereby enabling the RF repeaters to be conventional RF repeaters in Method A.

FIG. 2 is a block diagram of wireless communication network 100 illustrating one example of Method A. In this example, wireless communication paths 0, 1, and 2 are partially or completely blocked by a building 202, a tree 204,

and a building 206, respectively. MRA module 114 and/or MRA module 118 may analyze one or more measured conditions of wireless paths 0, 1, 2, and 3 and determine that path 3 is the best path to transfer data between wireless communication node 102 and wireless communication node 104, such as in response to wireless communication path 3 having a highest RSRP, a highest RSRQ, a smallest time advance, or a smallest percentage of wireless communication path utilization. In response thereto, MRA module 114 and/or MRA module 118 cause wireless base station 112 and/or UE device 116 to route data between wireless communication node 102 and wireless communication node 104 solely via wireless communication path 1, i.e., using channel Ch31, RF repeater 110, and Ch32.

Method B—Combine Copies of a Signal

Referring again to FIG. 1, in method B, multiple copies of a wireless communication signal, i.e., either a downlink wireless communication signal or an uplink wireless communication signal, are transmitted between wireless base station 112 and UE device 116 via different respective wireless communication paths. In contrast with Method A, multiple copies of a wireless communication signal may be used by wireless base station 112 and UE device 116 in Method B. In particular, MRA module 114 and/or MRA module 118 cause wireless base station 112 to combine copies of an uplink wireless communication signal it receives via two or more of wireless communication paths 0, 1, 2, and 3 to generate a combined wireless communication signal. Similarly, MRA module 114 and/or MRA module 118 cause UE device 116 to combine copies of a downlink wireless communication signal it receives via two or more of wireless communication paths 0, 1, 2, and 3 to generate a combined wireless communication signal. Each combined wireless communication signal may have a greater amplitude than any one of its constituent copies of the wireless signal communication signal, which advantageously helps achieve a high signal to noise ratio. In some embodiments, the copies of the wireless communication signal are combined using a soft combining process. For example, wireless base station 112 may sum complex values of received copies of an uplink wireless communication signal to generate a combined uplink wireless communication signal, and UE device 116 may sum complex values of received copies of a downlink wireless communication signal to generate a combined downlink wireless communication signal. Certain embodiments of Method B do not require coordination of RF repeater 106, RF repeater 108, or RF repeater 110, thereby enabling the RF repeaters to be conventional RF repeaters in Method B.

Some embodiments of MRA module 114 and/or MRA module 118 are configured to cause wireless base station 112 and UE device 116 to combine only a predetermined quantity of received copies of a wireless communication signal, or to combine only received copies of a wireless communication signal that meet one or more predetermined criteria. For example, in one embodiment, MRA module 114 and/or MRA module 118 are configured to cause wireless base station 112 or UE device 116 to combine the two received copies of a wireless communications signal having largest RSRP values, largest RSRQ values, or smallest time advance values. As another example, in another embodiment, MRA module 114 and/or MRA module 118 are configured to cause wireless base station 112 or UE device 116 to combine only copies of a wireless communication signal having RSRP values exceeding a threshold value, RSRQ values exceeding a threshold value, or time advance values that are below a threshold value.

FIG. 3 is a block diagram of wireless communication network 100 illustrating one example of Method B. In this example, wireless communication paths 0 and 1 are blocked by a building 302 and a tree 304, respectively. Consequently, wireless communication node 104 is capable of receiving copies of a downlink wireless communication signal solely via wireless communication paths 2 and 3. Accordingly, UE device 116 receives copies 306 and 308 of a common downlink wireless communication signal, and UE device 116 combines these two copies to generate a combined downlink wireless communication signal 310, as symbolically shown by text “306+308=310” in wireless communication node 104. The magnitude of combined downlink wireless communication signal 310 is greater than the respective magnitudes of either of copies 306 and 308 of the downlink wireless communication signal. Therefore, Method B increases signal to noise ratio of downlink wireless communication signals processed by UE device 116. Wireless base station 112 could analogously receive and combine copies of an uplink wireless communication sent via wireless communication paths 2 and 3, in the FIG. 3 example.

Method C—Joint Transmission

Referring again to FIG. 1, in Method C, MRA module 114 and MRA module 118 cooperate such that data is transferred between wireless communication node 102 and wireless communication node 104 using joint transmission, such that different downlink data and/or uplink data are transferred via two or more of wireless communication paths 0, 1, 2, and 3. For example, in particular embodiments, MRA modules 114 and 118 are configured such that wireless base station 112 sends different downlink data to each of RF repeaters 106, 108, and 110 via respective wireless communication signals in wireless channels Ch11, Ch21, and Ch31. RF repeaters 106, 108, and 110 then forward the received downlink data to UE device 116 via respective wireless communication signals in each of wireless channels Ch12, Ch22, and Ch32, and UE device 116 aggregates the received wireless communication signals. Accordingly, UE device 116 receives downlink data from wireless base station 112 via multiple wireless communication paths. Similarly, in certain embodiments, MRA modules 114 and 118 are configured such that UE device 116 sends different uplink data to each of RF repeaters 106, 108, and 110 via respective wireless communication signals in each of wireless channels Ch12, Ch22, and Ch32. RF repeaters 106, 108, and 110 then forward the received uplink data to wireless base station 112 via respective wireless communication signals in each of wireless channels Ch11, Ch21, and Ch31, and wireless base station 112 aggregates the received wireless communication signals. Accordingly, wireless base station 112 receives different uplink from UE device 116 via multiple wireless communication paths. Direct path 0, i.e., wireless channel Ch0, may be additionally be used in conjunction with one or more of wireless communication paths 1, 2, and 3 to further increase number of data transfer paths between wireless communication nodes 102 and 104.

Such ability to send different data via multiple wireless communication paths in Method C enables wireless communication network 100 to achieve a significantly larger throughput than would be feasible using conventional techniques. Additionally, the multiple wireless communication paths will typically be at different angles with respect to each of wireless base station 112 and UE device 116, such that there may be small correlation, or even essentially no correlation between wireless communication paths 0, 1, 2, and 3, which further increases maximum theoretical

throughput of wireless communication network 100. Furthermore, use of multiple wireless communication paths for data transmission in wireless communication network 100 helps mitigate shadowing, small-scale fading, and Doppler shift. Furthermore, certain embodiments of Method C do not require coordination of RF repeater 106, RF repeater 108, or RF repeater 110, thereby enabling the RF repeaters to be conventional RF repeaters.

FIG. 4 is a block diagram of wireless communication network 100 illustrating one example of Method C. In this example, wireless communication path 1 is blocked by a tree 402. However, wireless communication paths 0, 2, and 3 are available for data transmission between wireless communication nodes 102 and 104. Accordingly, MRA module 114 and MRA module 118 cooperate to provide for joint transmission of data between wireless base station 112 and UE device 116 using wireless communication paths 0, 2, and 3. For example MRA module 114 and MRA module 118 may cooperate such that wireless base station 112 sends different downlink wireless communication signals 404, 406, and 408 to UE device 116 in wireless communication via paths 0, 2, and 3. Specifically, in this example, wireless base station 112 sends downlink wireless communication signals 404 directly to UE device 116 using wireless communication path 0, and wireless base station 112 sends downlink wireless communication signals 406 and 408 via wireless channels Ch21 and Ch31 to RF repeater 108 and RF repeater 110, respectively. RF repeaters 108 and 110 then forward downlink wireless communication signals 406 and 408 to UE device 116, respectively, and UE device 116 aggregates received wireless communication signals 404, 406, and 408.

Method D—Dynamic Beam Configuration

Referring again to FIG. 1, in certain embodiments, each wireless each RF repeater 106, 108, and 110 may receive and transmit a plurality of beams. For example, wireless channel Ch11 between wireless base station 112 and RF repeater 106 may include a plurality of beams, and wireless channel Ch12 between RF repeater 106 and UE device 116 may also include a plurality of beams. As such, each of wireless communication paths 0, 1, 2, and 3 may include a plurality of possible beams by which wireless communication signals may be transmitted. In method D, MRA module 114 and/or MRA module 118 select a single wireless communication beam, or a single set of wireless communication beams, of a single wireless communication path 0, 1, 2, or 3 to transmit data between wireless communication node 102 and wireless communication node 104, such as based on one or more measured conditions of the wireless communication paths, to help optimize wireless communication between the two nodes. As such, method D is similar to method A, but method D further includes wireless communication beam coordination. The measured conditions of wireless communication paths in method D may include measured conditions of respective wireless communication beams of the wireless communication paths.

In certain embodiments of Method D, UE device 116 may measure conditions of each wireless communication beam received from each RF repeater 106, 108, and 110, and UE device 116 may send data representing the measured conditions of the wireless communication beams to MRA module 114 and/or MRA module 118. Similarly, wireless base station 112 may measure conditions of each wireless communication beam received from each RF repeater 106, 108, and 110, and wireless base station 112 may send data representing the measured conditions of the wireless communication beams to MRA module 114 and/or MRA module 118. Such measurements are performed, for example, on a

periodic basis, on an aperiodic basis, or in response to an event. MRA module 114 and/or MRA module 118 may then select a single wireless communication beam, or a single set of wireless communication beams, for transmitting data between wireless communication nodes 102 and 104, based at least in part on the data representing the measured conditions of the wireless communication beams. For example, MRA module 114 and/or MRA module 118 may select whichever wireless communication beam, or set of wireless communication beams, has a highest RSRP, a highest RSRQ, a smallest time advance, a smallest percentage of channel utilization, etc. MRA module 114 and/or MRA module 118 then cause data to be routed between wireless communication node 102 and wireless communication node 104 solely via the one selected wireless communication beam, or selected set of wireless communication beams, such as by controlling wireless base station 112 and/or UE device 116 to route data accordingly.

In particular embodiments, wireless base station 112 and/or UE device 116 may not distinguish between wireless communication beams sent directly between the two devices and wireless communication beams sent between the two devices via one or more RF repeaters 106, 108, and 110. Additionally, some embodiments of Method D do not require coordination of RF repeater 106, RF repeater 108, or RF repeater 110, thereby enabling the RF repeaters to be conventional RF repeaters.

Method E—Beam Direction

Method E is an alternate embodiment of Method B further including wireless communication beam coordination. In Method E, multiple copies of a downlink wireless communication signal or an uplink wireless communication signal are sent by wireless base station 112 or UE device 116 via different respective wireless communication beams. For example, wireless base station 112 may send four copies of a common downlink wireless communication signal via one or more wireless communication beams along each of wireless communication paths 0, 1, 2, and 3. As another example, UE device 116 may send four copies of an uplink wireless communication signal via one or more wireless communication beams along each of wireless communication paths 0, 1, 2, and 3. Each RF repeater 106, 108, and 110 amplifies and forwards each wireless communication beam that it receives. As such, if all four wireless communication paths 0, 1, 2, and 3 are available, UE device 116 will receive four copies of a common downlink wireless communication signal via four wireless communication beams. Similarly, if all four wireless communication paths 0, 1, 2, and 3 are available, wireless base station 112 will receive four copies of a common uplink wireless communication signal via four wireless communication beams. MRA module 114 and/or MRA module 118 cause wireless base station 112 to combine copies of an uplink wireless communication signal it receives via two or more wireless communication beams. Additionally, MRA module 114 and/or MRA module 118 cause UE device 116 to combine copies of a downlink wireless communication signal it receives via two or more wireless communication beams.

Method F—Multiple Streams

In particular embodiments of wireless communication network 100, wireless each RF repeater 106, 108, and 110 may receive and transmit a plurality of beams, and each beam may include a plurality of streams. Method F is similar to method C, but joint transmission is extended across multiple streams in each of two or more of wireless communication paths 0, 1, 2, and 3. For example, assume that each of wireless communication paths 0, 1, 2, and 3 is

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available, and that each path supports two streams between wireless communication nodes **102** and **104**. In this example, there are eight routes between wireless communication nodes **102** and **104**, which significantly increases throughput of wireless communication network **100** compared to a scenario where joint transmission is not used. Additionally, in a manner analogous to that discussed above with respect to Method C, these multiple wireless communication paths enabled by Method F will typically be at different angles with respect to each of wireless base station **112** and UE device **116**, such that there may be small correlation, or even essentially no correlation between paths **0**, **1**, **2**, and **3**, which further increases maximum theoretical throughput of wireless communication network **100**. Furthermore, use of multiple wireless communication paths for data transmission in wireless communication network **100** helps mitigate shadowing, small-scale fading, and Doppler shift.

Method G—Frequency/Time Resource Aggregation

Methods A-F, discussed above, advantageously implement spatial diversity of communication resources to promote high performance of wireless communication network **100**. Any of methods A-F can be modified to further implement frequency and/or time diversity of communication resources among wireless communication paths **0**, **1**, **2**, and **3**, among wireless communication beams of a given wireless communication path, and/or among streams of a given wireless communication beam. Such implementation of frequency and/or time diversity in wireless communication network **100** may further promote high throughput and high reliability.

FIG. **5** is a graph **500** of frequency versus time illustrating one example of how frequency and time diversity may be implemented by MRA module **114** and/or MRA module **118**. In the FIG. **5** example, a respective physical resource block (PRB) **502**, **504**, **506**, and **508** is allocated to each of wireless communication path **0**, wireless communication path **1**, wireless communication path **2**, and wireless communication path **3**. Each PRB specifies a combination of frequency and time that may be used for the path. For example, wireless base station **112**, RF repeater **106**, and UE devices **116** are permitted to collectively use solely the frequency and time range specified by PRB **504** to implement wireless communication path **1**.

Method H—Fake Base Station Detection

Method H detects a fake wireless base station. In method H, MRA module **114** and/or MRA module **118** analyze wireless communication signals received at a node in wireless communication network **100**, such as wireless communication signals received by UE device **116** or wireless communication signals received by wireless base station **112**, to determine angle of incidence of the received wireless communication signals. For example, MRA module **114** and/or MRA module **118** may determine angle of incidence of wireless communication signals received by UE device **116**. Reception of a plurality of wireless communication signals at substantially different angles of incidence indicates that the sending wireless base station is likely a legitimate wireless base station. In contrast, reception of a plurality of wireless communication signals at substantially common angles of incidence indicates that the sending wireless base station is likely a fake wireless base station. Accordingly, particular embodiments of MRA module **114** and/or MRA module **118** are configured to determine that a particular wireless base station is fake in response to angle of incidence of received wireless communication signals from the wireless base station differing by less than a

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predetermined threshold value. Additionally, some embodiment of MRA module **114** and/or MRA module **118** are configured to determine that a particular wireless base station is legitimate in response to angle of incidence of received wireless communication signals from the wireless base station differing by more than a predetermined threshold value.

Method I—Dynamic Power Control

In certain embodiments of wireless communication network **100**, amplification of RF repeaters **106**, **108**, and **110** may be adjusted, and method I includes MRA module **114** and/or MRA module **118** adjusting amplification or “power” of one or more of RF repeaters **106**, **108**, and **110** to optimize or more characteristics of wireless communication network **100**. Method I may be incorporated with any of the methods discussed above. For example, method I may be incorporated with method B to help further increase signal to noise ratio of a sum of copies of a downlink or uplink wireless communication signal.

Load Balancing

A conventional smart RF repeater is capable of receiving data from a donor wireless communication node via a first wireless communication beam and forwarding the data to a child using a second wireless communication beam, in a first slot. The conventional smart RF repeater may be configured such that there is a semi-static linkage between the first wireless communication beam and the second wireless communication beam, and the second wireless communication beam may be dependent on the first slot. Accordingly, a conventional smart repeater is ill-suited for handling aggregated data for one or more target wireless communication nodes over different wireless communication beams. For example, assume a hypothetical scenario where a conventional smart RF repeater receives the following wireless communication signals from a donor: (a) a first wireless communication beam including data **1** for forwarding to a first UE device (UE **1**) and (b) a second wireless communication beam including data **2** for forwarding to a second UE device (UE **2**). A conventional smart RF repeater configured with a semi-static linkage between wireless communication beams would be incapable of determining how to forward data **1** and data **2** to its intended UE device.

Applicant has developed new systems and methods for load balancing which at least partially overcome the above discussed drawbacks of conventional smart repeaters. Certain embodiments are configured to associate frequency regions, or bandwidth parts (BWPs), within a given wireless communication beam with respective destination wireless communication nodes, for forwarding data. As such, a recipient smart RF repeater, or other recipient network element, can determine where to forward data carried by a wireless communication beam at least partially based on a BWP that the data occupies in the wireless communication beam. Accordingly, particular embodiments of the new systems and methods are capable of receiving data for multiple child wireless communication nodes, e.g., UE devices, customer premises equipment (CPE), and other smart RF repeaters, and forwarding the data to the child wireless communication nodes over different respective wireless communication beams.

FIG. **6** is a block diagram of a wireless communication network **600** including an embodiment of the new systems for load balancing. Wireless communication network **600** includes a wireless communication node **602**, a wireless communication node **604**, a wireless communication node **606**, a wireless communication node **608**, a smart RF repeater **610**, a smart RF repeater **612**, and a smart RF

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repeater **614**. The aforesaid elements of wireless communication network **600** as configured, for example, in a mesh configuration. The number of wireless communication nodes in wireless communication network **600**, as well as the number of smart repeaters in wireless communication network **600**, may vary without departing from the scope hereof.

Each wireless communication node **602**, **604**, **606**, and **608** is configured to transmit and/or receive wireless communication beams. In some embodiments, each wireless communication node **602**, **604**, **606**, and **608** includes one or more of a wireless base station (e.g., a 3GPP wireless base station, a Wi-Fi wireless base station, a satellite wireless base station, a Bluetooth wireless base station, a LoRa wireless base station, a Zigbee wireless base station, a Z-Wave wireless base station, etc), CPE, a mobile telephone, a computer, a set-top device, a data storage device, an IoT device, an entertainment device, a computer networking device, a smartwatch, a wearable device with wireless capability, a medical device, a security device, and a monitoring device. Each wireless communication node **602**, **604**, **606**, and **608** need not have the same configuration.

Each smart RF repeater **610**, **612**, and **614** is configured to receive data via wireless communication beams and forward data carried by the wireless communication beams to child wireless communication nodes according to respective BWPs associated with the data. Smart RF repeaters **610**, **612**, and **614** could be replaced with other devices capable of forwarding wireless communication beams, e.g., IABs, as long as the other devices are capable of forwarding data carried by wireless communication beams to child nodes according to respective BWPs associated with the data. Additionally, one or more of smart repeaters **610**, **612**, and **614** could be combined with another device, such as a wireless base station, CPE, or a UE device.

FIG. 6 depicts an example of operation of wireless communication network **600** in load balancing where data is being transmitted from wireless communication node **602** to other nodes of wireless communication network **600**, such that wireless communication node **602** is acting as a “donor” wireless communication node. It is understood, though, that wireless communication network **600** could operate with other nodes being donor wireless communication nodes, as well as with wireless communication beams taking paths other than those illustrated in the FIG. 6 example.

In the FIG. 6 example, wireless communication node **602** transmits wireless communication beams **616** and **618** to smart RF repeaters **610** and **612**, respectively, such as in different respective time slots. Each of wireless communication beams **616** and **618** is structured, in the frequency domain, into different BWPs, where each BWP encompasses a respective portion of a wireless communication channel bandwidth of the beam. In particular, wireless communication beam **616** is structured into bandwidth parts BWP01, BWP031, and BWP04, which collectively encompass a bandwidth Ch BW01 of a wireless communication channel associated with wireless communication beam **616**, as illustrated in FIG. 6. Each of bandwidth parts BWP01, BWP031, and BWP04 includes data associated with a different respective wireless communication node of wireless communication network **600**. Specifically, bandwidth part BWP01 includes data associated with smart RF repeater **610**, bandwidth part BWP031 includes data associated with smart RF repeater **614**, and bandwidth part BWP04 includes data associated with wireless communication node **604**. Wireless communication beam **618**, in turn, is structured into bandwidth parts BWP02, BWP032, and BWP05, which

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collectively encompass a bandwidth Ch BW02 of a wireless communication channel associated with wireless communication beam **618**, as illustrated in FIG. 6. While not required, in certain embodiments, bandwidth Ch BW01 is equal to bandwidth Ch BW02. Each of bandwidth parts BWP02, BWP032, and BWP05 also includes data associated with a different respective wireless communication node of wireless communication network **600**. Specifically, bandwidth part BWP02 includes data associated with smart RF repeater **612**, bandwidth part BWP032 includes data associated with smart RF repeater **614**, and bandwidth part BWP05 includes data associated with wireless communication node **606**.

Smart RF repeater **610** receives a message associating bandwidth parts of wireless communication beam **616** with destination wireless communication nodes. Such message may be sent, for example, via data associated with bandwidth part BWP01, which is destined for smart RF repeater **610**. The message may associate, for example, (a) bandwidth part BWP031 with smart RF repeater **614** and (b) bandwidth part BWP04 with wireless communication node **604**. In some embodiments, the message is radio resource control (RRC) message. Smart RF repeater **610** then forwards data associated with each of bandwidth parts BWP031 and BWP04 to its intended destination according to destinations specified in the received message. For example, smart RF repeater **610** may receive wireless communication beam **616** in a first time slot, and smart RF repeater **610** may then forward data associated with bandwidth part BWP04 to wireless communication node **604** in a second time slot via a wireless communication beam **620**, and smart RF repeater **610** may forward data associated with bandwidth part BWP031 to smart RF repeater **614** in a third time slot via a wireless communication beam **622**.

Smart RF repeater **612** receives a message associating bandwidth parts of wireless communication beam **618** with destination wireless communication nodes. Such message may be sent, for example, via data associated with bandwidth part BWP02, which is destined for smart RF repeater **612**. The message may associate, for example, (a) bandwidth part BWP032 with smart RF repeater **614** and (b) bandwidth part BWP05 with wireless communication node **606**. In some embodiments, the message is a RRC message. Smart RF repeater **612** then forwards data associated with each of bandwidth part BWP032 and bandwidth part BWP05 to its intended destination according to destinations specified in the received message. For example, smart RF repeater **612** may receive wireless communication beam **618** in a fourth time slot, and smart RF repeater **612** may then forward data associated with bandwidth part BWP05 to wireless communication node **606** in a fifth time slot via a wireless communication beam **624**, and smart RF repeater **612** may route data associated with bandwidth part BWP032 to smart RF repeater **614** in a sixth time slot via a wireless communication beam **626**.

Smart RF repeater **614** aggregates (a) data associated with bandwidth part BWP031 received via wireless communication beam **622**, and (b) data associated with bandwidth part BWP032 received via wireless communication beam **626**, to yield aggregated data associated with bandwidth part BWP03, as illustrated in FIG. 6. In some embodiments, bandwidth part BWP03 has a bandwidth Ch BW03 which is greater than either of bandwidth Ch BW01 or Ch BW02. In some embodiments, smart RF repeater **614** receives each of wireless communication beams **622** and **626** in different respective time slots. Smart repeater **614** then forwards aggregated data associated with bandwidth part BWP03 to wireless communication node **608** via a wireless communi-

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cation beam **628**. However, in some alternate embodiments, smart RF repeater **614** is the final destination of aggregated data associated with bandwidth part BWP**03**. Additionally, in some alternate embodiments, smart repeater **614** is replaced with another device, such as CPE, a UE device, or a wireless base station, that is capable of aggregating data associated with of bandwidth part BWP**031** and bandwidth part BWP**032** to yield aggregated data associated with bandwidth part BWP**03**.

In some embodiments, each smart RF repeater **610**, **612**, and **614** forwards received data that is destined for another wireless communication node without processing the data. Accordingly, in particular embodiments, error management messages, e.g., error correction messages and/or error control messages, are transported from an origination wireless communication node to a destination wireless communication node via one or more smart RF repeaters **610**, **612**, **614** without processing the error correction messages, thereby achieving end-to-end error correction and/or error control. Examples of such error correction messages include, but are not limited to, hybrid automatic repeat request (HARQ) messages, such as HARQ ACK messages and/or HARQ NACK messages. Additionally, some embodiments of wireless communication network **600** are configured to retransmit a data structure, such as a transport block (TB), via an alternate path in the event that the data structure, as received by the destination node, includes one or more errors.

For instance, FIG. 7 is a block diagram of a wireless communication network **700**, which is one embodiment of wireless communication network **600** configured to retransmit a data structure via an alternate path in response to the data structure including errors. Wireless communication nodes **604**, **606**, and **608** of FIG. 6 are not shown in FIG. 7 for illustrative clarity. FIG. 7 depicts an example where a transport block included in bandwidth part BWP**031** includes an error. In particular, wireless communication node **602** transmits a wireless communication beam **702** to smart RF repeater **610**, where wireless communication beam **702** includes bandwidth parts BWP**01**, BWP**031**, and BWP**04**. Smart RF repeater **610** then forwards data associated with bandwidth part BWP**031** to smart RF repeater **614** via a wireless communication beam **704**. Bandwidth part BWP**031** includes a plurality of transport blocks, although only one transport block, i.e., transport block TBx**01**, is shown in FIG. 7 for illustrative clarity. Each transport block includes a plurality of coding block groups (CBGs) (not shown), and each coding block group includes a plurality of coding blocks (CBs) (not shown).

Smart RF repeater **614** processes each coding block of each coding block group in each transport block, to determine if there are any errors. Additionally, smart RF repeater **614** sends a HARQ ACK message to wireless communication node **602** via smart RF repeater **610** and/or **612** in response to each data unit that is successfully received by smart repeater **614**, and smart RF repeater **614** sends a HARQ NACK message to wireless communication node **602** via smart RF repeater **610** and/or **612** in response to each data unit that is not successfully received by smart repeater **614**. As such, wireless communication network **700** supports end-to-end HARQ between wireless communication nodes **602** and **604** and across one or more smart RF repeaters.

In the FIG. 7 example, smart repeater **614** determines that one coding block in a first copy **705** of transport block TBx**01** includes an error. Specifically, coding block CB-**031-x02** includes an error as symbolically shown in FIG. 7 by the coding block being filled with cross-hatching, while coding

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blocks CB-**031-x01** and CB-**031-x03** do not include errors, as symbolically shown in FIG. 7 by the coding block being filled with light shading. Smart repeater **614** then sends a message to wireless communication node **602** advising of the error in transport block TBx**01**, within a prescribed time interval associated with coding block CB-**031-x02**.

Wireless communication node **602** retransmits transport block TBx**01** to smart RF repeater **614** via a different path than that used to originally send transport block TBx**01** to smart RF repeater **614**, in response to receiving the message indicating the error in transport block TBx**01**. In particular, wireless communication node **602** transmits a wireless communication beam **706** to smart RF repeater **612**, with transport block TBx**01** appended to the end of bandwidth part BWP**0321**. Smart RF repeater **612** then forwards data associated with BWP**0321** to smart RF repeater **614** via a wireless communication beam **704**, such that smart RF repeater **614** receives a second copy **710** of transport block TBx**01** via bandwidth part BWP**0321**.

Coding blocks CB-**031-x02** and CB-**031-x03** in second copy **710** of transport block TBx**01** are error free. However, coding block CB-**031-x01** in second copy **710** of transport block TBx**01** includes an error. Nevertheless, smart RF repeater **614** has received error-free copies of each coding blocks CB-**031-x01**, CB-**031-x02**, and CB-**031-x03**, between first copy **705** and second copy **710** of transport block TBx**01**. Accordingly, smart RF repeater **614** generates a combined copy **712** of transport block TBx**01** from error-free coding blocks of each of first copy **705** of TBx**01** and second copy of TBx**01**, using, for example, a chase combining process. While the FIG. 7 example illustrates combined copy **712** transport block of TBx**01** being formed from only two copies of the transport block, combined copy **712** of transport block of TBx**01** could be formed using additional copies of the transport block, such as by executing a chase combining process a plurality of times.

Variable Time Division Duplex Data Allocation

Time division duplex (TDD) may be used to transmit uplink and downlink data in a communication network. In TDD, a common frequency band is used to transmit both uplink data and downlink data, but the uplink data and the downlink data are transmitted at different times, e.g., in different slots. Applicant has developed systems and methods for variable TDD data allocation that significantly advance the state of the art. For example, in certain embodiments, an on-demand portion of TDD communication resources is allocated to a communication node having a highest data transmission requirement. The same allocation may be replicated by a smart RF repeater (or other device) on subsequent communication links, such as between the smart repeater and one or more downstream child communication nodes, to minimize (or eliminate) the need for physical (PHY) layer processing and media access control (MAC) processing in the smart RF repeater. Furthermore, in particular embodiments, unused TDD allocations may be used by a smart RF repeater for local data transmission. Moreover, in some embodiments, unused TDD allocations may be used for direct communication between two nodes in a communication network.

FIG. 8 is a block diagram of a wireless communication network **800** including an embodiment of the new systems for variable TDD data allocation. Wireless communication network **800** includes a wireless communication node **802**, a wireless communication node **804**, a wireless communication node **806**, a wireless communication node **808**, and a network controller **810**. Wireless communication nodes **802**, **804**, **806**, and **808** are configured, for example, in a mesh

configuration. Wireless communication nodes **804**, **806**, and **808** are downstream of wireless communication node **802**, and wireless communication node **802** is accordingly upstream of wireless communication nodes **804**, **806**, and **808**. The number of wireless communication nodes in wireless communication network **800** may vary without departing from the scope hereof.

Wireless communication node **808** is a smart RF repeater that is configured to receive data frames carried by wireless communication beams and forward the data frames via other wireless communication beams. Wireless communication node **808** could alternately be another type of RF repeating device, such as an IAB, having capability to support the disclosed methods for variable TDD data allocation.

Each wireless communication node **802**, **804**, and **806** is configured to transmit and/or receive wireless communication beams. In some embodiments, each wireless communication node **802**, **804**, and **806** includes one or more of a wireless base station (e.g., a 3GPP wireless base station, a Wi-Fi wireless base station, a satellite wireless base station, a Bluetooth wireless base station, a LoRa wireless base station, a Zigbee wireless base station, a Z-Wave wireless base station, etc.), CPE, a mobile telephone, a computer, a set-top device, a data storage device, an IoT device, an entertainment device, a computer networking device, a smartwatch, a wearable device with wireless capability, a medical device, a security device, a monitoring device, a RF repeater, a smart RF repeater, and an IAB. Each wireless communication node **802**, **804**, and **806** need not have the same configuration. For example, in certain embodiments, wireless communication node **802** is a wireless base station (e.g., a 3GPP wireless base station such as a LTE wireless base station, a 5G wireless base station, or a 6G wireless base station), wireless communication node **804** is a smart RF repeater or CPE, and wireless communication node **806** is a smart RF repeater or CPE.

Network controller **810** is configured to control one or more of wireless communication nodes **802**, **804**, **806**, and **808** to implement variable TDD data allocation in wireless communication network **800**. Some embodiments of network controller **810** include a processor (not shown) and a memory (not shown), where the processor is configured to execute instructions in the form of software and/or firmware stored in the memory to implement variable TDD data allocation in wireless communication network **800**. While network controller **810** is depicted as being a discrete element, network controller **810** could be integrated with one or more other elements. For, example, in some embodiments, network controller **810** is at least partially integrated with a core wireless communication network (not shown), such as a 3GPP core wireless communication network, a satellite core wireless communication network, or a Wi-Fi core wireless communication network. As another example, in some embodiments, network controller **810** is at least partially integrated within one or more of wireless communication nodes **802**, **804**, **806**, and **808**. Network controller **810** may be a distributed controller, e.g., network controller **810** may be distributed among two or more elements, such as among a core wireless communication network and one or more of wireless communication nodes **802**, **804**, **806**, and **808**, or among two or more of wireless communication nodes **802**, **804**, **806**, and **808**.

FIG. 8 depicts an example of operation of wireless communication network **800** where data is being transmitted between wireless communication node **802** and wireless communication nodes **804**, **806**, and **808** in a point to multi-point manner, such that wireless communication node

802 is acting as a donor wireless communication node and each wireless communication node **804**, **806**, and **808** is acting as a child wireless communication node. It is understood, though, that wireless communication network **800** could operate with another node being a donor wireless communication node and with other nodes being child wireless communication nodes, as well as with wireless communication beams taking paths other than those illustrated in the FIG. 8.

In the FIG. 8 example, data is transmitted between wireless communication node **802** and wireless communication node **808** via first TDD data frames **812** carried by one or more first wireless communication beams **814**. Additionally, data is transmitted between wireless communication node **808** and wireless communication node **804** via second data TDD frames **824** carried by one or more second wireless communication beams **826**, and data is transferred between wireless communication node **808** and wireless communication node **806** via third TDD data frames **828** carried by one or more third wireless communication beams **830**.

Data carried by wireless communication beams in wireless communication network **800** is structured in the frequency domain into different BWPs, where each BWP encompasses a respective portion of a wireless communication channel bandwidth between two wireless communication nodes. In particular, each wireless communication beam is structured into bandwidth parts BWP01, BWP03, and BWP04, which collectively encompass a bandwidth Ch BW of a wireless communication channel between two wireless communication nodes. A bandwidth part may be used for downlink and uplink data transmission at different respective times, or a bandwidth part may be used to transmit data between different wireless communication nodes at different respective times. In the FIG. 8 example, bandwidth part BWP01 is referred to as BWP011 when used for downlink data transmission to wireless communication node **808**, and bandwidth part BWP01 is referred to as BWP012 when used for downlink data transmission to wireless communication node **806**. Bandwidth part BWP03 is referred to as BWP031 when used for downlink data transmission to wireless communication node **804**, and bandwidth part BWP03 is referred to as BWP032 when used for uplink data transmission from wireless communication node **804**. Bandwidth part BWP04 is referred to as BWP041 when used for uplink data transmission from wireless communication node **808**, and bandwidth part BWP04 is referred to as BWP042 when used for uplink data transmission from wireless communication node **806**. The quantity and use of BWPs in wireless communication network **800** may vary without departing from the scope hereof.

Network controller **810** is configured to allocate data in first TDD data frames **812** according to a first data frame allocation **816**. First data frame allocation **816** includes a fixed downlink (DL) portion **818**, a fixed uplink (UL) portion **820**, and an on-demand downlink/uplink (DL/UL) portion **822**, where the portions of first data frame allocation **816** are separated from each other in the time domain. In some embodiments, first data frames **812** at least partially conform to a 3GPP wireless communication protocol, including but not limited to a LTE wireless communication protocol, a 5G wireless communication protocol, a 6G wireless communication protocol, extensions of any of the foregoing wireless communication protocols, or successors to any of the foregoing wireless communication protocols.

Network controller **810** is further configured to variably allocate on-demand portion **822** to data to be transferred to

one or more client wireless communication nodes, such as based on current operating conditions of wireless communication network **800**. For example, network controller **810** may be configured to determine a highest traffic child wireless communication node of child wireless communication nodes **804**, **806**, and **808**, where the highest traffic child wireless communication node is a child wireless communication node **804**, **806**, or **808** having a highest data transmission requirement, e.g. a child wireless communication node requiring a highest uplink bandwidth and/or a highest downlink bandwidth. Network controller **810** may then allocate on-demand portion **822** of first data frame allocation **816** to data associated with the highest traffic child wireless communication node, while allocating fixed portions **818** and **820** of first data frame allocation **816** to data associated with other child wireless communication nodes. Network controller **810** may additionally variably allocate symbols of on-demand DL/UL portion **822** to data associated with the highest traffic child wireless communication node, such as by changing a split between downlink data and uplink data according to changing data transmission requirements of the highest traffic child wireless communication node.

The FIG. **8** example assumes that wireless communication node **804** currently has a highest data transmission requirement and is therefore the current highest traffic child wireless communication node. Accordingly, first data frame allocation **816** depicts on-demand DL/UL portion **822** being allocated to BWP**031** and BWP**032**, i.e., to both downlink data and uplink data associated with wireless communication node **804**. Fixed DL portion **818** is allocated to BWP**011** and BWP**012**, and fixed UL portion **820** is allocated to BWP**041** and BWP**042**, in the FIG. **8** example.

Wireless communication node **808** forwards received data to another wireless communication node according to (a) the BWP including the data and (b) the portion of a TDD data frame including the data. In particular, wireless communication node **808** forward data received via first data frames **812** as follows: (a) downlink data included in BWP**031** of on-demand DL/UL portion **822** is forwarded to wireless communication node **804** via second TDD data frames **824** carried by second wireless communication beams **826**, and (b) downlink data included in BWP**012** of fixed DL portion **818** is forwarded to wireless communication node **806** via third TDD data frames **828** carried by third wireless communication beams **830**. Wireless communication node **808** does not forward downlink data included in BWP**011** of fixed DL portion **818** received via first data frames **812** because wireless communication node **808** is the ultimate destination for this downlink data. Additionally, wireless communication node **808** forwards uplink data included in BWP**032** of on-demand DL/UL portion **822** and received via second TDD data frames **824** to wireless communication node **802**, and wireless communication node **808** forwards uplink data included in BWP**042** of fixed UL portion **820** and received via third data frames **828** to wireless communication node **802**. Wireless communication node **808** additionally sends internally generated uplink data to wireless communication node **802** via BWP**041** of fixed UL portion **820**.

In particular embodiments, network controller **810** replicates first data frame allocation **816** on second TDD data frames **824** and third TDD data frames **828**, which helps minimize PHY layer processing requirements and MAC layer signaling of requirements wireless communication node **808**. For example, FIG. **8** illustrates that data is allocated in second data frames **824** and in third data frames

828 according to a second data frame allocation **832** and a third data frame allocation **834**, respectively. Second data frame allocation **832** is a replica of first data frame allocation **816** in that non-empty portions of second data frame allocation **832** are allocated to bandwidth parts in the same manner as in first data frame allocation **816**. For example, on-demand DL/UL portion **822** is allocated to BWP**031** and BWP**032** in second data frame allocation **832** in the same manner as in first data frame allocation **816**. Additionally, third data frame allocation **834** is a replica of first data frame allocation **816** in that non-empty portions of third data frame allocation **834** are allocated to bandwidth parts in the same manner as in first data frame allocation **816**. For example, BWP**012** and BWP**042** are respectively allocated to fixed DL portion **818** and fixed uplink portion **820**, respectively, in third data frame allocation **834** in the same manner as in first data frame allocation **816**.

Certain embodiments of wireless communication node **808** are configured to forward local data between client wireless communication nodes, as well as forward data between donor wireless communication node **802** and client wireless communication nodes **804** and **806**. For example, some embodiments of wireless communication node **808** are configured to (a) prioritize forwarding of data frames between wireless communication node **802** and client wireless communication nodes **804** and **806** and (b) forward data frames between wireless communication nodes **804** and **806** using spare communication resources in wireless communication network **800**, e.g., using empty portions of TDD data frames and/or unused RF spectrum.

For example, FIG. **9** is a block diagram of a wireless communication network **900**, which is one embodiment of wireless communication network **800** where wireless communication node **808** is configured to forward local data between client wireless communication nodes **804** and **806**, as well as forward data between donor wireless communication node **802** and client wireless communication nodes **804** and **806**. FIG. **9** illustrates an example of operation of wireless communication network **900** where wireless communication node **808** relays local data between wireless communication node **804** and wireless communication node **806** using empty portions of TDD data frames. FIG. **9** assumes that data is not currently being transmitted between wireless communication node **802** and wireless communication node **804**, and on-demand DL/UL portion **822** of TDD data frames is therefore unused.

Data is transmitted between wireless communication node **804** and wireless communication node **806** via fourth TDD data frames **924** carried by one or more wireless communication beams **926**, where data is allocated in fourth TDD data frames **924** according to a fourth data frame allocation **932**. As illustrated in FIG. **9**, fourth data frame allocation **932** allocates on-demand DL/UL portion **822** to BWP**031** and BWP**032** for local downlink data transmission and local uplink data transmission, respectively, between wireless communication node **804** and wireless communication node **808**. Additionally, data is transmitted between wireless communication node **806** and wireless communication node **808** via fifth TDD data frames **928** carried by one or more wireless communication beams **930**, where data is allocated in fifth TDD data frames **928** according to a fifth data frame allocation **934**. Fifth data frame allocation **934** allocates on-demand DL/UL portion **822** to BWP**031** and BWP**032** for local downlink data transmission and local uplink data transmission, respectively, between wireless communication node **808** and wireless communication node **806**. In some embodiments, data is transferred between

wireless communication node **804** and wireless communication node **806** using a device-to-device communication protocol, e.g., using a 3GPP Sidelink wireless communication protocol.

Referring again to FIG. 8, in certain embodiments of wireless communication network **800**, two or more child wireless communication nodes are configured to directly communicate with each other using spare communication resources in wireless communication network **800**, e.g., using empty portions of TDD data frames and/or unused RF spectrum. For example, FIG. 10 is a block diagram of a wireless communication network **1000**, which is one embodiment of wireless communication network **800** where wireless communication nodes **804** and **806** are configured to directly communicate with each other. FIG. 10 assumes that data is not currently being transmitted between wireless communication node **802** and wireless communication node **804**, and on-demand DL/UL portion **822** of TDD data frames is therefore unused.

Data is directly transmitted between wireless communication node **804** and wireless communication node **806** via sixth TDD data frames **1036** carried by one or more wireless communication beams **1038**, where data is allocated in sixth TDD data frames **1036** according to a sixth data frame allocation **1040**. As illustrated in FIG. 9, sixth data frame allocation **1040** allocates on-demand DL/UL portion **822** to BWP031 and BWP032 for direct local downlink data transmission and local uplink data transmission, respectively, between wireless communication node **808** and wireless communication node **806**. In some embodiments, data is transferred between wireless communication node **804** and wireless communication node **806** using a device-to-device communication protocol, e.g., using a 3GPP Sidelink communication protocol.

Referring again to FIG. 8, in certain embodiments of wireless communication network **800**, network controller **810** is further configured to supplement data transmission between wireless communication nodes using spare resources in wireless communication network **800**, e.g., using empty portions of TDD data frames and/or unused RF spectrum. For example, consider a scenario where no data is currently being transmitted between donor wireless communication node **802** and child wireless communication node **806**. Portions of TDD data frames allocated to data associated with wireless communication node **806**, i.e. BWP012 of fixed DL portion **818** and BWP042 of fixed UL portion **820**, could temporarily be used to supplement data transfer between other nodes in wireless communication network **800**, e.g., to supplement data transfer between donor wireless communication node **802** and child wireless communication node **804**, such as to increase data transmission capacity between donor wireless communication node **802** and child wireless communication node **804**.

Furthermore, in some embodiments of wireless communication network **800**, network controller **810** is further configured to support data transmission between wireless communication nodes using spare resources in wireless communication network **800**, e.g., using empty portions of TDD data frames and/or unused RF spectrum, in the event of failure of a communication link using allocated resources. For example, consider a scenario where (a) no data is currently being transmitted between donor wireless communication node **802** and child wireless communication node **806** and (b) a communication link between donor wireless communication node **802** and child wireless communication node **804** using allocated on-demand DL/UL portion **822** of TDD data frames fails, such as due to interference. Portions

of TDD data frames allocated to data associated with wireless communication node **806**, i.e. BWP012 of fixed DL portion **818** and BWP042 of fixed UL portion **820**, could temporarily be used to transfer data between donor wireless communication node **802** and child wireless communication node **804**, thereby providing redundancy in wireless communication network **800** in the event of failure of a primary communication link between donor wireless communication node **802** and child wireless communication node **804**.

Coreset Reuse

A coreset is a set of physical resources used to carry control information in a wireless communication network. For example, a coreset in a 3GPP wireless communication network is a set of resources, e.g., physical resource blocks, within a specific area of a downlink resource grid, and the coreset in a 3GPP wireless communication network carries, for example, physical downlink control channel (PDCCH) information. Applicant has developed systems and methods for coreset reuse which use a common coreset across two or more communication links in a wireless communication network, thereby helping minimize resources required to generate and process coresets.

FIG. 11 is a block diagram of a wireless communication network **1100**, which is one embodiment of a wireless communication network including a system for coreset reuse. Wireless communication network **1100** includes a wireless communication node **1102**, a wireless communication node **1104**, a wireless communication node **1106**, and a wireless communication node **1108**. Wireless communication nodes **1102**, **1104**, **1106**, and **1108** are configured, for example, in a mesh configuration. Wireless communication nodes **1104**, **1106**, and **1108** are downstream of wireless communication node **1102**, and wireless communication node **1102** is accordingly upstream of wireless communication nodes **1104**, **1106**, and **1108**. The number of wireless communication nodes in wireless communication network **1100** may vary without departing from the scope hereof.

Wireless communication node **1108** is a smart RF repeater that is configured to receive data frames carried by wireless communication beams and forward the data frames via other wireless communication beams. Wireless communication node **1108** could be configured to operate with either analog wireless communication beams or digital wireless communication beams. Wireless communication node **1108** could alternately be another type of RF repeating device, such as an IAB, having capability to support the disclosed methods for variable TDD data allocation.

Each wireless communication node **1102**, **1104**, and **1106** is configured to transmit and/or receive wireless communication beams. In some embodiments, each wireless communication node **1102**, **1104**, and **1106** includes one or more of a wireless base station (e.g., a 3GPP wireless base station, a Wi-Fi wireless base station, a satellite wireless base station, a Bluetooth wireless base station, a LoRa wireless base station, a Zigbee wireless base station, a Z-Wave wireless base station, etc.), CPE, a mobile telephone, a computer, a set-top device, a data storage device, an IoT device, an entertainment device, a computer networking device, a smartwatch, a wearable device with wireless capability, a medical device, a security device, a monitoring device, a RF repeater, a smart RF repeater, and an IAB. Each wireless communication node **1102**, **1104**, and **1106** need not have the same configuration. For example, in certain embodiments, wireless communication node **1102** is a wireless base station (e.g., a 3GPP wireless base station such as a LTE wireless base station, a 5G wireless base station, or a 6G wireless base station), wireless communication node **1104** is

a smart RF repeater or CPE, and wireless communication node **1106** is a smart RF repeater or CPE.

FIG. **11** depicts an example of operation of wireless communication network **1100** where data is being transmitted between wireless communication node **1102** and wireless communication nodes **1104**, **1106**, and **1108** in a point to multi-point manner, such that wireless communication node **1102** is acting as a donor wireless communication node and each wireless communication node **1104**, **1106**, and **1108** is acting as a child wireless communication node. It is understood, though, that wireless communication network **1100** could operate with another node being a donor wireless communication node and with other nodes being child wireless communication nodes, as well as with wireless communication beams taking paths other than those illustrated in the FIG. **11**.

Data is transmitted between wireless communication node **1102** and wireless communication node **1108** via one or more first wireless communication beams **1110**. Additionally, data is transmitted between wireless communication node **1108** and wireless communication node **1104** via one or more second wireless communication beams **1112**, and data is transmitted between wireless communication node **1108** and wireless communication node **1106** via one or more third wireless communication beams **1114**. First wireless communication beams **1110**, second wireless communication beams **1112**, and third wireless communication beams **1114** may be either analog wireless communication beams or digital wireless communication beams. Data carried by wireless communication beams in wireless communication network **1100** is structured in the frequency domain into different BWPs, where each BWP encompasses a respective portion of a wireless communication channel bandwidth between two wireless communication nodes. In particular, each wireless communication beam is structured into bandwidth parts BWP01, BWP02, and BWP03, which collectively encompass a bandwidth Ch BW of a wireless communication channel between two wireless communication nodes, as illustrated in FIG. **11**. Each of bandwidth parts BWP01, BWP02, and BWP03 includes data associated with a different respective child wireless communication node of wireless communication network **1100**. Specifically, bandwidth part BWP01 includes data associated with wireless communication node **1108**, bandwidth part BWP02 includes data associated with wireless communication node **1104**, and bandwidth part BWP03 includes data associated with wireless communication node **1106**. The number and use of BWPs in wireless communication network **1100** may vary without departing from the scope hereof.

First wireless communication beams **1110** include a respective coreset associated with each wireless communication node downstream of wireless communication node **1102**. In particular, first wireless communication beams **1110** include a first coreset CS01 associated with wireless communication node **1108**, a second coreset CS02 associated with wireless communication node **1104**, and a third coreset CS03 associated with wireless communication node **1106**. In some embodiments, each of first coreset CS01, second coreset CS02, and third coreset CS03 is a 3GPP wireless communication coreset and includes, for example, PDCCH information.

Wireless communication node **1108** exchanges data received via a wireless communication beam according to a bandwidth part of the beam. For example, wireless communication node **1108** forwards data of BWP02 received via one or more first wireless communication beams **1110** to wireless communication node **1104** via one or more second

wireless communication beams **1112**. As another example, wireless communication node **1108** forwards data of BWP03 received via one or more first wireless communication beams **1110** to wireless communication node **1106** via one or more third wireless communication beams **1114**. As another example, wireless communication node **1108** does not forward data of BWP01 received via one or more first wireless communication beams **1110** because wireless communication node **1108** is the final destination for this data.

Importantly, wireless communication node **1108** is configured to reuse coresets, which reduces processing requirements of wireless communication node **1108**. For example, wireless communication node **1108** generates second wireless communication beams **1112** such that they contain the same coreset associated with wireless communication node **1104** as first wireless communication beams **1110**, or in other words, so that first wireless communication beams **1110** and second wireless communication beams **1112** include a common coreset CS02 associated with wireless communication node **1104**. As another example, wireless communication node **1108** generates third wireless communication beams **1114** such that they contain the same coreset associated with wireless communication node **1106** as first wireless communication beams **1110**, or in other words, so that first wireless communication beams **1110** and third wireless communication beams **1114** include a common coreset CS03 associated with wireless communication node **1106**.

In some embodiments, wireless communication node **1108** forwards received data that is destined for another wireless communication node without processing the data. Accordingly, in particular embodiments, wireless communication node **1108** forwards error management messages, e.g., error correction messages and/or error control messages, from an origination wireless communication node to a destination wireless communication node without processing the error correction messages, thereby enabling end-to-end error correction and/or error control in wireless communication network **1100**. For example, assume that wireless communication node **1108** receives an error management message from wireless communication node **1104**. Certain embodiments of wireless communication node **1108** forward the error management message to wireless communication node **1102** without processing the error management message. As another example, assume that wireless communication node **1108** receives an error management message from wireless communication node **1102**. Certain embodiments of wireless communication node **1108** may forward the error management message to wireless communication node **1104** or to wireless communication node **1106** without processing the error management message. Examples of error management messages include, but are not limited to, HARQ messages, such as HARQ ACK messages and/or HARQ NACK messages.

Bandwidth Part Coding

Applicant has developed systems and methods for BWP coding which may reduce or eliminate BWP reprocessing at a smart RF repeater or similar device. For example, FIG. **12** is a block diagram of a wireless communication network **1200**, which is one embodiment of a wireless communication network including a system for BWP coding. Wireless communication network **1200** includes a wireless communication node **1202**, a wireless communication node **1204**, a wireless communication node **1206**, and a wireless communication node **1208**. Wireless communication nodes **1202**, **1204**, **1206**, and **1208** are configured, for example, in a mesh configuration. Wireless communication nodes **1204**, **1206**, and **1208** are downstream of wireless communication node

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1202, and wireless communication node 1202 is accordingly upstream of wireless communication nodes 1204, 1206, and 1208. The number of wireless communication nodes in wireless communication network 1200 may vary without departing from the scope hereof.

Wireless communication node 1208 is a smart RF repeater that is configured to receive data carried by wireless communication beams and forward the data via other wireless communication beams. Wireless communication node 1208 could alternately be another type of RF repeating device, such as an IAB, having capability to support the disclosed methods for BWP coding.

Wireless communication node 1208 is capable of recoding wireless communication data transport blocks. For example, assume that wireless communication node 1208 receives data via incoming data transport blocks carried by an incoming wireless communication beam and that wireless communication node 1208 forwards the data via outgoing data transport blocks carried by an outgoing wireless communication beam. Wireless communication node 1208 is capable of coding the outgoing data transport blocks using a different modulation coding scheme (MCS) than was used to code the incoming data transport blocks, such as in response to the outgoing wireless communication beam being unable to support as high of a modulation coding scheme order as the incoming wireless communication beam.

Each wireless communication node 1202, 1204, and 1206 is configured to transmit and/or receive wireless communication beams. In some embodiments, each wireless communication node 1202, 1204, and 1206 includes one or more of a wireless base station (e.g., a 3GPP wireless base station, a Wi-Fi wireless base station, a satellite wireless base station, a Bluetooth wireless base station, a LoRa wireless base station, a Zigbee wireless base station, a Z-Wave wireless base station, etc.), CPE, a mobile telephone, a computer, a set-top device, a data storage device, an IoT device, an entertainment device, a computer networking device, a smartwatch, a wearable device with wireless capability, a medical device, a security device, a monitoring device, a RF repeater, a smart RF repeater, and an IAB. Each wireless communication node 1202, 1204, and 1206 need not have the same configuration. For example, in certain embodiments, wireless communication node 1202 is a wireless base station (e.g., a 3GPP wireless base station such as a LTE wireless base station, a 5G wireless base station, or a 6G wireless base station), wireless communication node 1204 is a smart RF repeater or CPE, and wireless communication node 1206 is a smart RF repeater or CPE.

FIG. 12 depicts an example of operation of wireless communication network 1200 where data is being transmitted between wireless communication node 1202 and wireless communication nodes 1204, 1206, and 1208 in a point to multi-point manner, such that wireless communication node 1202 is acting as a donor wireless communication node and each wireless communication node 1204, 1206, and 1208 is acting as a child wireless communication node. It is understood, though, that wireless communication network 1200 could operate with another node being a donor wireless communication node and with other nodes being child wireless communication nodes, as well as with wireless communication beams taking paths other than those illustrated in the FIG. 12.

Data is transmitted between wireless communication node 1202 and wireless communication node 1208 via one or more first wireless communication beams 1210. Additionally, data is transmitted between wireless communication

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node 1208 and wireless communication node 1204 via one or more second wireless communication beams 1212, and data is transmitted between wireless communication node 1208 and wireless communication node 1206 via one or more third wireless communication beams 1214. Data transmitted by wireless communication beams 1210, 1212, and 1214 is structured in the frequency domain into different BWPs, where each BWP encompasses a respective portion of a wireless communication channel bandwidth between two wireless communication nodes. In particular, each wireless communication beam is structured into bandwidth parts BWP01, BWP02, and BWP03, which collectively encompass a bandwidth Ch BW of a wireless communication channel between two wireless communication nodes. Each of bandwidth parts BWP01, BWP01, and BWP03 includes data associated with a different respective child wireless communication node of wireless communication network 1200. Specifically, bandwidth part BWP01 includes first data associated with wireless communication node 1208, bandwidth part BWP02 includes second data associated with wireless communication node 1204, and bandwidth part BWP03 includes third data associated with wireless communication node 1206. The number and use of BWPs in wireless communication network 1200 may vary without departing from the scope hereof. For example, in some alternate embodiments, data is transmitted via solely a single BWP, such as a single BWP that occupies an entirety of a respective slot.

Wireless communication node 1208 exchanges data received via a wireless communication beam according to a bandwidth part of the beam. For example, wireless communication node 1208 forwards second data of BWP02 received via one or more first wireless communication beams 1210 to wireless communication node 1204 via one or more second wireless communication beams 1212. As another example, wireless communication node 1208 forwards third data of BWP03 received via one or more first wireless communications beams 1210 to wireless communication node 1206 via one or more third wireless communication beams 1214. As another example, wireless communication node 1208 does not forward first data of BWP01 received via one or more first wireless communications beams 1210 because wireless communication node 1208 is the final destination for this data.

FIG. 12 depicts two paths including wireless communication node 1208. A first path 1216 transfers data between wireless communication node 1202 and wireless communication node 1204, and first path 1216 includes first wireless communication beams 1210, wireless communication node 1208, and second wireless communication beams 1212. A second path 1218 transfers data between wireless communication node 1202 and wireless communication node 1206, and second path 1218 includes first wireless communication beams 1210, wireless communication node 1208, and third wireless communication beams 1214.

In view of wireless communication node 1208 being capable of recoding data transport blocks, a given path 1216 or 1218 may including data transport blocks coded using different modulation coding schemes. For example, assume that first wireless communication beams 1210 are relatively free of interference while second wireless communication beams 1212 are subjected to significant interference. Wireless communication node 1202 may code data transport blocks for transmission via first wireless communication beams 1210 using a high order modulation coding scheme given that first wireless communication beams 1210 are relatively free of interference. However, wireless commu-

nication node **1208** may recode data transport blocks for transmission via second wireless communication beams **1212** using a low order modulation coding scheme given that second wireless communication beams **1212** are subject to significant interference. In such case, first path **1216** would include transport blocks code using two different modulation coding scheme, and wireless communication node **1208** would accordingly recode data transport blocks, i.e., code data transport blocks using a different modulation coding scheme than that was used to originally code the data transport blocks.

Wireless communication network is configured to implement BWP coding to reduce, or even eliminate, the need for wireless communication node **1208** recode data transport blocks to compensate for unfavorable wireless communication conditions downstream of wireless communication node **1208**. In particular, wireless communication node **1202** is configured to code data transport blocks of a given BWP at a lowest modulation coding scheme used in a path including the BWP, to prevent the possible need for wireless communication node **1208** to recode the data transport blocks at a lower order modulation coding scheme. For example, assume that (a) wireless communication node **1202** codes data transport blocks of BWP02 using a first modulation coding scheme for transmission to wireless communication node **1208** via first wireless communication beams **1210**, and (b) wireless communication node **1208** recodes data transport blocks of BWP02 using a second modulation coding scheme for transmission to wireless communication node **1204** via second wireless communication beams **1212**, where the second modulation coding scheme has a lower order than the first modulation coding scheme. Wireless communication node **1208** may change it operation to code data transport blocks of BWP02 for transmission via first wireless communication beams **1210** using the lowest order modulation coding scheme of path **1216**, i.e., the second modulation coding scheme, to eliminate the need for wireless communication node **1208** to recode data transport blocks using the second modulation coding scheme.

FIG. **13** is a flow chart of method **1300** for BWP coding, which is one example of possible operation of wireless communication network **1300**. It is understood, however, that wireless communication network **1300** is not limited to operating according to method **1300**. Additionally, method **1300** could be applied to wireless communication networks other than wireless communication network **1300**.

In a block **1302** of method **1300** donor wireless communication node **1202** obtains a plurality of metrics, where each metric represents communication quality a first BWP of a respective wireless communication beam in wireless communication network **1200**. In one example of block **1302**, (a) the first BWP is BWP02, and the first BWP accordingly encompasses an identical frequency range in each of wireless communication beams **1210** and **1212**, (b) each metric is an interference plus noise ratio (SINR) of BWP02 of its respective wireless communication beam, (c) donor wireless communication node **1202** measures SINR of one or more data transport blocks of BWP02 in one or more first wireless communication beams **1210**, henceforth referred to as SINR-1, (c) child wireless communication node **1208** measures SINR of one or more data transport blocks of BWP02 in one or more second wireless communication beams **1212**, henceforth referred to SINR-2, and (d) child wireless communication node **1208** forwards SINR-2 to donor wireless communication node **1202**. In some embodiments, SINR-1 and SINR-2 are measured using common data transport

blocks in each of first wireless communication beams **1210** and second wireless communication beams **1212**.

In a block **1304** of method **1300**, donor wireless communication node **1202** determines a lowest metric of the plurality of metrics. In one example of block **1304**, donor wireless communication node determines that SINR-2 is lower than SINR-1, such that SINR-2 is a lowest metric. In a block **1306** of method **1306**, donor wireless communication node **1202** codes data transport blocks of the first BWP using a modulation coding scheme used to code data transport blocks of the first BWP of a wireless communication beam corresponding to the lowest metric. In one example of block **1306**, donor wireless communication node **1202** codes data transport blocks of BWP02 for transmission via first wireless communication beams **1210** using the same modulation coding scheme used by child wireless communication node **1208** to code transport data blocks for transmission via second wireless communication beams **1212**.

Combinations of Features

Features described above may be combined in various ways without departing from the scope hereof. The following examples illustrate some possible combinations.

(A1) A method for multiple repeater aggregation includes (1) receiving, at a first node of a wireless communication network, wireless communication signals from a second node of the wireless communication network via a plurality of wireless communication paths, each of the plurality of wireless communication paths including a respective radio frequency (RF) repeater, and (2) selecting one of the plurality of wireless communication paths for use in transferring data between the first node of the wireless communication network and the second node of the wireless communication network, at least partially based on measured conditions of the plurality of wireless communication paths.

(A2) The method denoted as (A1) may further include causing data to be transferred between the first node of the wireless communication network and the second node of the wireless communication network via the selected one of the plurality of wireless communication paths, without coordinating with the respective RF repeaters of the plurality of wireless communication paths.

(A3) In either one of the methods denoted as (A1) and (A2), the first node of the wireless communication network may include a user equipment (UE) device, and the second node of the wireless communication network may include a wireless base station.

(A4) In either one of the methods denoted as (A1) and (A2), the first node of the wireless communication network may include a wireless base station, and the second node of the wireless communication network may include a user equipment (UE) device.

(A5) In any one of the methods denoted as (A1) through (A4), the measured conditions of the plurality of wireless communication paths may include one or more of a reference signal received power (RSRP) of each wireless communication path of the plurality of wireless communication paths, a reference signal received quality (RSRQ) of each wireless communication path of the plurality of wireless communication paths, a time advance of each wireless communication path of the plurality of wireless communication paths, a cell identifier (ID) of each wireless communication path of the plurality of wireless communication paths, a wireless communication beam identifier (ID) of each wireless communication path of the plurality of wireless communication paths, and a percentage of wireless

channel utilization of each wireless communication path of the plurality of wireless communication paths.

(A6) In any one of the methods denoted as (A1) through (A5), selecting one of the plurality of wireless communication paths to use for transferring data between the first node of the wireless communication network and the second node of the wireless communication network may include selecting one or more wireless communication beams of the selected one of the plurality of wireless communication paths.

(B1) A method for multiple repeater aggregation includes (1) receiving a plurality of copies of a wireless communication signal, each copy of the wireless communication signal being received via a respective wireless communication path of a plurality of wireless communication paths, each of the plurality of wireless communication paths including a respective radio frequency (RF) repeater, and (2) combining the plurality of copies of the wireless communication signal to generate a combined wireless communication signal.

(B2) In the method denoted as (B1), combining the plurality of copies of the wireless communication signal to generate the combined wireless communication signal may include soft combining the plurality of copies of the wireless communication signal.

(B3) In either one of the methods denoted as (B1) and (B2), the steps of receiving and combining may be performed at a user equipment device.

(B4) In either one of the methods denoted as (B1) and (B2), the steps of receiving and combining may be performed at a wireless base station.

(B5) In either one of the methods denoted as (B1) through (B4), the wireless communication signal may include one or more of downlink data and uplink data.

(B6) Any one of the methods denoted as (B1) through (B5) may further include receiving each copy of the wireless communication signal via a respective wireless communication beam.

(C1) A method for multiple repeater aggregation includes (1) receiving, at a first node of a wireless communication network, a plurality of wireless communication signals from a second node of the wireless communication network via a plurality of wireless communication paths, each of the plurality of wireless communication signals representing different data, and each of the plurality of wireless communication paths including a respective radio frequency (RF) repeater, and (2) aggregating the received plurality of wireless communication signals at the first node.

(C2) The method denoted as (C1) may further include causing data to be transferred between the first node of the wireless communication network and the second node of the wireless communication network via the plurality of wireless communication paths, without coordinating with the respective RF repeaters of the plurality of wireless communication paths.

(C3) In either one of the methods denoted as (C1) and (C2), the first node of the wireless communication network may include a wireless base station, and the second node of the wireless communication network may include a user equipment (UE) device.

(C4) In either one of the methods denoted as (C1) and (C2), the first node of the wireless communication network may include a user equipment (UE) device, and the second node of the wireless communication network may include a wireless base station.

(C5) In any one of the methods denoted as (C1) through (C4), at least two wireless communication signals of the

plurality of wireless communication signals may be part of a common wireless communication beam.

(C6) In any one of the methods denoted as (C1) through (C5), at least two wireless communication signals of the plurality of wireless communication signals may be within different respective frequencies ranges and/or may be within different respective time ranges.

(D1) A method for load balancing, includes receiving, from a donor wireless communication node, a message, the message associating (a) a first bandwidth part with a first destination wireless communication node, and (b) a second bandwidth part with a second destination wireless communication node.

(D2) The method denoted as (D1) may further include (1) receiving, in a first time slot, data associated with the first bandwidth part, and (2) forwarding the data associated with the first bandwidth part to the first destination wireless communication node, in a second time slot.

(D3) The method denoted as (D2) may further include (1) receiving, in the first time slot, data associated with the second bandwidth part, and (2) forwarding the data associated with the second bandwidth part to the second destination wireless communication node, in a third time slot.

(D4) In any one of the methods denoted as (D1) through (D3), the donor wireless communication node may be a wireless base station.

(E1) A method for load balancing includes (1) receiving, via a first wireless communication beam, data associated with a first bandwidth part of the first wireless communication beam, (2) receiving, via a second wireless communication beam, data associated with a first bandwidth part of the second wireless communication beam, and (3) aggregating (a) the data associated with the first bandwidth part of the first wireless communication beam and (b) the data associated with the first bandwidth part of the second wireless communication beam.

(E2) The method denoted as (E1) may further include receiving the data associated with the first bandwidth part of the first wireless communication beam and the data associated with the first bandwidth part of the second wireless communication beam in different respective time slots.

(E3) Either one of the methods denoted as (E1) and (E2) may further include (1) receiving the data associated with the first bandwidth part of the first wireless communication beam from a first radio frequency (RF) repeater, and (2) receiving the data associated with the first bandwidth part of the second wireless communication beam from a second radio frequency (RF) repeater.

(F1) A method for end-to-end error management in a wireless communication network includes (1) receiving, at a first wireless communication node, data from a second wireless communication node via a first radio-frequency (RF) repeater, and (2) sending one or more error management messages associated with the data from the first wireless communication node to the second wireless communication node via one or more RF repeaters.

(F2) In the method denoted as (F1), the one or more error management messages may include one or more hybrid automatic repeat request (HARQ) messages.

(F3) In either one of the methods denoted as (F1) and (F2), the data received via the first RF repeater may include one or more errors, and the method may further include receiving a copy of the data via a second RF repeater.

(F4) The method denoted as (F3) may further include generating a combined copy of the data from (a) the data received via the first RF repeater and (b) the copy of the data received via the second RF repeater.

(F5) In the method denoted as (F4), generating the combined copy of the data may include using a chase combining process.

(G1) A method for variable time division duplex (TDD) data allocation in a communication network including a donor wireless communication node and a plurality of child wireless communication nodes includes (1) determining a highest traffic child wireless communication node of the plurality of child wireless communication nodes, the highest traffic child wireless communication node having a highest data transmission requirement of the plurality of child wireless communication nodes, and (2) allocating data in first TDD data frames to be transmitted between the donor wireless communication node and a first child wireless communication node of the plurality of child wireless communication nodes according to a first data frame allocation, the first data frame allocation including allocating an on-demand portion of the first TDD data frames to data associated with the highest traffic child wireless communication node.

(G2) In the method denoted as (G1), the highest traffic child wireless communication node and the first child wireless communication node may be the same child wireless communication node.

(G3) In the method denoted as (G1), the highest traffic child wireless communication node and the first child wireless communication node may be different respective wireless communication nodes.

(G4) Any one of the methods denoted as (G1) through (G4) may further include replicating the first data frame allocation on second TDD data frames to be transmitted between the first child wireless communication node and a second child wireless communication node of the plurality of child wireless communication nodes.

(G5) In the method denoted as (G4), the first child wireless communication node may be a first smart radio frequency (RF) repeater, and the second child wireless communication node may be one of a second smart RF repeater and customer premises equipment (CPE).

(G6) In any one of the methods denoted as (G1) through (G5), the first TDD data frames may at least partially conform to a Third Generation Partnership Project (3GPP) wireless communication protocol.

(G7) In the method denoted as (G6), the 3GPP wireless communication protocol may be one of a long term evolution (LTE) wireless communication protocol, a fifth generation (5G) wireless communication protocol, and a sixth generation (6G) wireless communication protocol.

(G8) Any one of the methods denoted as (G1) through (G7) may further include transmitting data between the first child wireless communication node and a second wireless communication node at least partially using unused portions of the first data frame allocation.

(G9) Any one of the methods denoted as (G1) through (G7) may further include transmitting data between (a) a second child wireless communication node of the plurality of child wireless communication nodes and (b) a third child wireless communication node of the plurality of child wireless communication nodes, at least partially using unused portions of the first data frame allocation.

(I1) A method for coreset reuse in a wireless communication network includes (1) receiving, at a first wireless communication node of the wireless communication network, first data via one or more first wireless communication beams, the first data including a first coreset, and (2) forwarding, at the first wireless communication node, the first data to a second wireless communication node via one

or more second wireless communication beams, the one or more first wireless communication beams and the one or more second wireless communication beams carrying a common coreset.

(I2) In the method denoted as (I1), the common coreset may be a Third Generation Partnership Project (3GPP) coreset.

(I3) In any one of the methods denoted as (I1) and (I2), the common coreset may include physical downlink control channel (PDCCH) information.

(I4) In any one of the messages denoted as (I1) through (I3), the first wireless communication node may be a smart radio frequency (RF) repeater.

(I5) Any one of the methods denoted as (I1) through (I3) may further include (1) receiving, at the first wireless communication node, one or more error management messages associated with the first data, and (2) forwarding the one or more error management messages to another wireless communication node without processing the one or more error management messages.

(I6) In the method denoted as (I5), the one or more error management messages may include one or more hybrid automatic repeat request (HARQ) messages.

(I7) In any one of the methods denoted as (I1) through (I6), the one or more first wireless communication beams and the one or more second wireless communication beams may be either analog wireless communication beams or digital wireless communication beams.

(J1) A method for bandwidth part (BWP) coding in a wireless communication network includes (1) determining a plurality of metrics, each metric representing communication quality of a first BWP of a respective wireless communication beam in the wireless communication network, (2) determining a lowest metric of the plurality of metrics, and (3) coding, at a donor wireless communication node of the wireless communication network, data transport blocks of the first BWP using a same modulation coding scheme used to code data transport blocks of the first BWP of a wireless communication beam corresponding to the lowest metric.

(J2) In the method denoted as (J1), each metric of the plurality of metrics may represent a signal to interference plus noise ratio (SINR) of the first BWP of its respective wireless communication beam.

(J3) In any one of the methods denoted as (J1) and (J2), the first BWP of each wireless communication beam of the plurality of wireless communication beams may encompass an identical frequency range.

(J4) In any one of the methods denoted as (J1) through (J3), the first BWP of each wireless communication beam may occupy an entirety of a respective slot

Changes may be made in the above methods, devices, and systems without departing from the scope hereof. It should thus be noted that the matter contained in the above description and shown in the accompanying drawings should be interpreted as illustrative and not in a limiting sense. The following claims are intended to cover generic and specific features described herein, as well as all statements of the scope of the present method and system, which as a matter of language, might be said to fall therebetween.

What is claimed is:

1. A method for variable time division duplex (TDD) data allocation in a communication network including a donor wireless communication node and a plurality of child wireless communication nodes, the method comprising:
determining a highest traffic child wireless communication node of the plurality of child wireless communication nodes,

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the highest traffic child wireless communication node having a highest data transmission requirement of the plurality of child wireless communication nodes; allocating data in first TDD data frames to be transmitted between the donor wireless communication node and a first child wireless communication node of the plurality of child wireless communication nodes according to a first data frame allocation,

the first data frame allocation including allocating an on-demand portion of the first TDD data frames to data associated with the highest traffic child wireless communication node; and

transmitting data between

(a) a second child wireless communication node of the plurality of child wireless communication nodes and
(b) a third child wireless communication node of the plurality of child wireless communication nodes, at least partially using unused portions of the first data frame allocation.

2. The method of claim 1, wherein the highest traffic child wireless communication node and the first child wireless communication node are the same child wireless communication node.

3. The method of claim 1, wherein the highest traffic child wireless communication node and the first child wireless communication node are different respective wireless communication nodes.

4. The method of claim 1, further comprising replicating the first data frame allocation on second TDD data frames to be transmitted between the first child wireless communication node and a fourth child wireless communication node of the plurality of child wireless communication nodes.

5. The method of claim 4, wherein:

the first child wireless communication node is a first smart radio frequency (RF) repeater, and

the fourth child wireless communication node is one of a second smart RF repeater and customer premises equipment (CPE).

6. The method of claim 1, wherein the first TDD data frames at least partially conform to a Third Generation Partnership Project (3GPP) wireless communication protocol.

7. The method of claim 6, wherein the 3GPP wireless communication protocol is one of a long term evolution (LTE) wireless communication protocol, a fifth generation (5G) wireless communication protocol, and a sixth generation (6G) wireless communication protocol.

8. The method of claim 1, further comprising transmitting data between the first child wireless communication node and a fourth wireless communication node at least partially using additional unused portions of the first data frame allocation.

9. A method for variable time division duplex (TDD) data allocation in a communication network including a donor wireless communication node and a plurality of child wireless communication nodes, the method comprising:

prioritizing use of first TDD data frames for transmitting data between the donor wireless communication node and the plurality of child wireless communication nodes;

determining that an on-demand portion of the first TDD data frames is not being used to transfer data between the donor wireless communication node and

a first child wireless communication node of the plurality of child wireless communication nodes;

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in response to determining that the on-demand portion of the first TDD data frames is not being used to transfer data between the donor wireless communication node and

the first child wireless communication node,

transmitting first data between the first child wireless communication node and

a second child wireless communication node of the plurality of child wireless communication nodes using the on-demand portion of first TDD data frames; and transmitting second data between

(a) a third child wireless communication node of the plurality of child wireless communication nodes and

(b) a fourth child wireless communication node of the plurality of child wireless communication nodes, at least partially using unused portions of a data frame allocation of the first TDD data frames.

10. The method of claim 9, further comprising transmitting the first data between the first child wireless communication node and the second child wireless communication node using the on-demand portion of the first TDD data frames via a fifth child wireless communication node of the plurality of child wireless communication nodes.

11. The method of claim 10, wherein the fifth child wireless communication node is a smart radio frequency (RF) repeater.

12. The method of claim 10, wherein the fifth child wireless communication node is an Integrated Access Backhaul (IAB) device.

13. The method of claim 9, wherein the first TDD data frames at least partially conform to a Third Generation Partnership Project (3GPP) wireless communication protocol.

14. The method of claim 9, wherein the first data is transferred between first child wireless communication node and the second child wireless communication node using the on-demand portion of the first TDD data frames without use of an intervening child wireless communication node of the plurality of child wireless communication nodes.

15. A method for variable time division duplex (TDD) data allocation in a communication network including a donor wireless communication node and a plurality of child wireless communication nodes, the method comprising:

allocating data in first TDD data frames to be transmitted between the donor wireless communication node and the plurality of child wireless communication nodes according to a first data frame allocation,

the first data frame allocation including

(i) an allocation of an on-demand portion of the first TDD data frames to data associated with a first child wireless communication node of the plurality of child wireless communication nodes and

(ii) an allocation of a first fixed portion of the first TDD data frames to data associated with a second child wireless communication node of the plurality of child wireless communication nodes;

detecting failure of a communication link between the donor wireless communication node and

the first child wireless communication node using the on-demand portion of the first TDD data frames;

in response to detecting the failure of the communication link between the donor wireless communication node and

the first child wireless communication node using the on-demand portion of the first TDD data frames,

temporarily using the first fixed portion of the first TDD data frames to transfer data between the donor wireless communication node and the first child wireless communication node; and

transmitting data between

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(a) a third child wireless communication node of the plurality of child wireless communication nodes and

(b) a fourth child wireless communication node of the plurality of child wireless communication nodes, at least partially using unused portions of the first data frame allocation.

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16. The method of claim **15**, further comprising transferring data between the donor wireless communication node and the first child wireless communication node via a fifth child wireless communication node of the plurality of child wireless communication nodes.

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17. The method of claim **16**, wherein the fifth child wireless communication node is a smart radio frequency (RF) repeater.

18. The method of claim **16**, wherein the fifth child wireless communication node is an Integrated Access Backhaul (IAB) device.

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19. The method of claim **15**, wherein the first TDD data frames at least partially conform to a Third Generation Partnership Project (3GPP) wireless communication protocol.

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